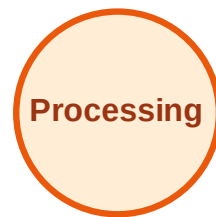


BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Queue: A

Visited: (none)



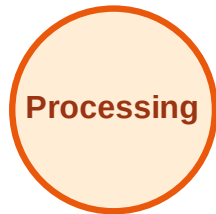
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



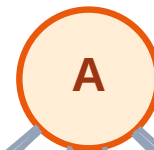
Discovered



Unseen

Queue: (empty)

Visited: (none)



A



B



C



G



K
(TARGET)



M



L



I



J



F



O



E



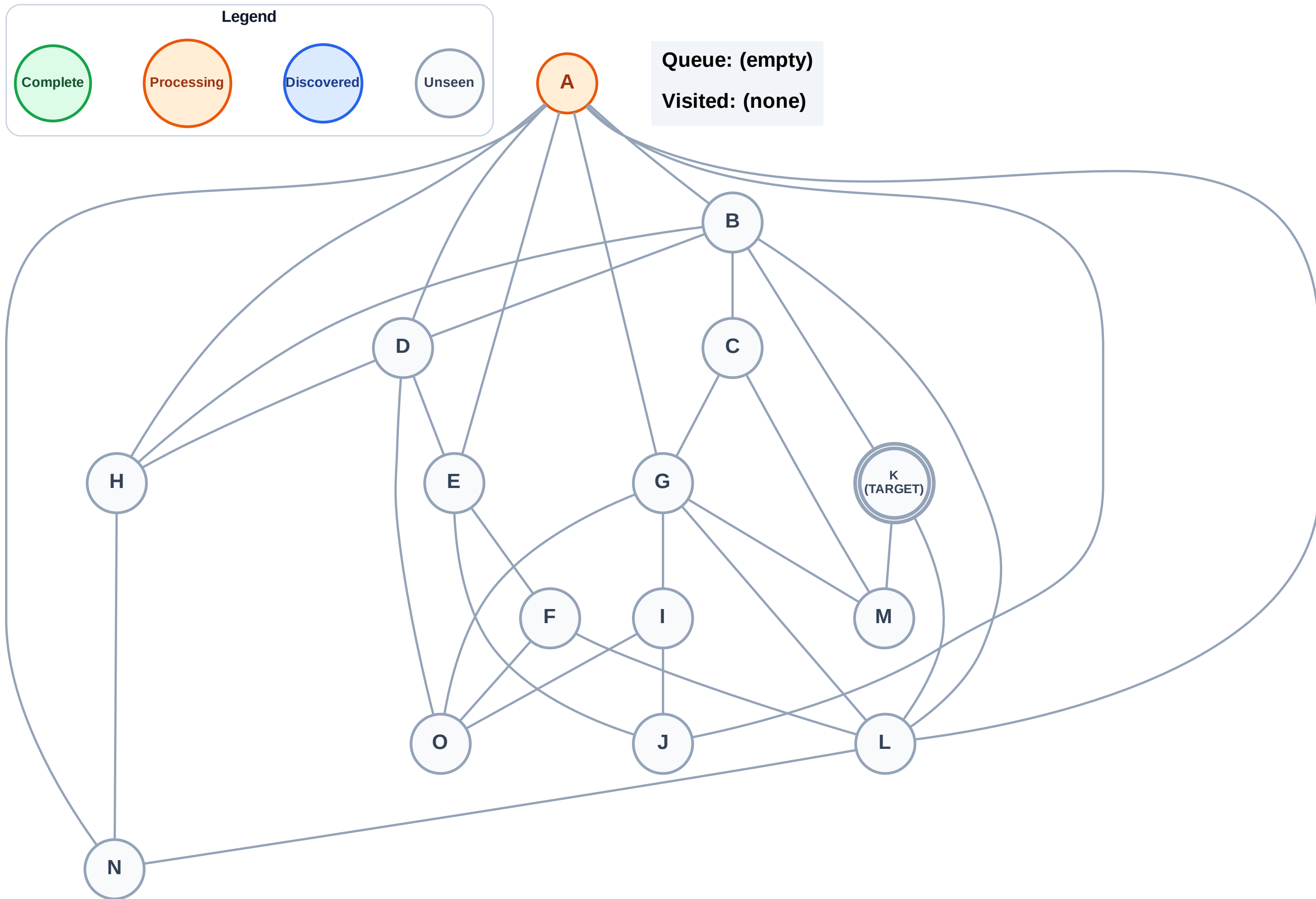
D



H



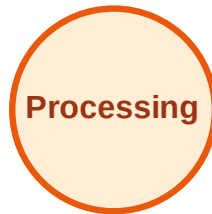
N



BFS Traversal

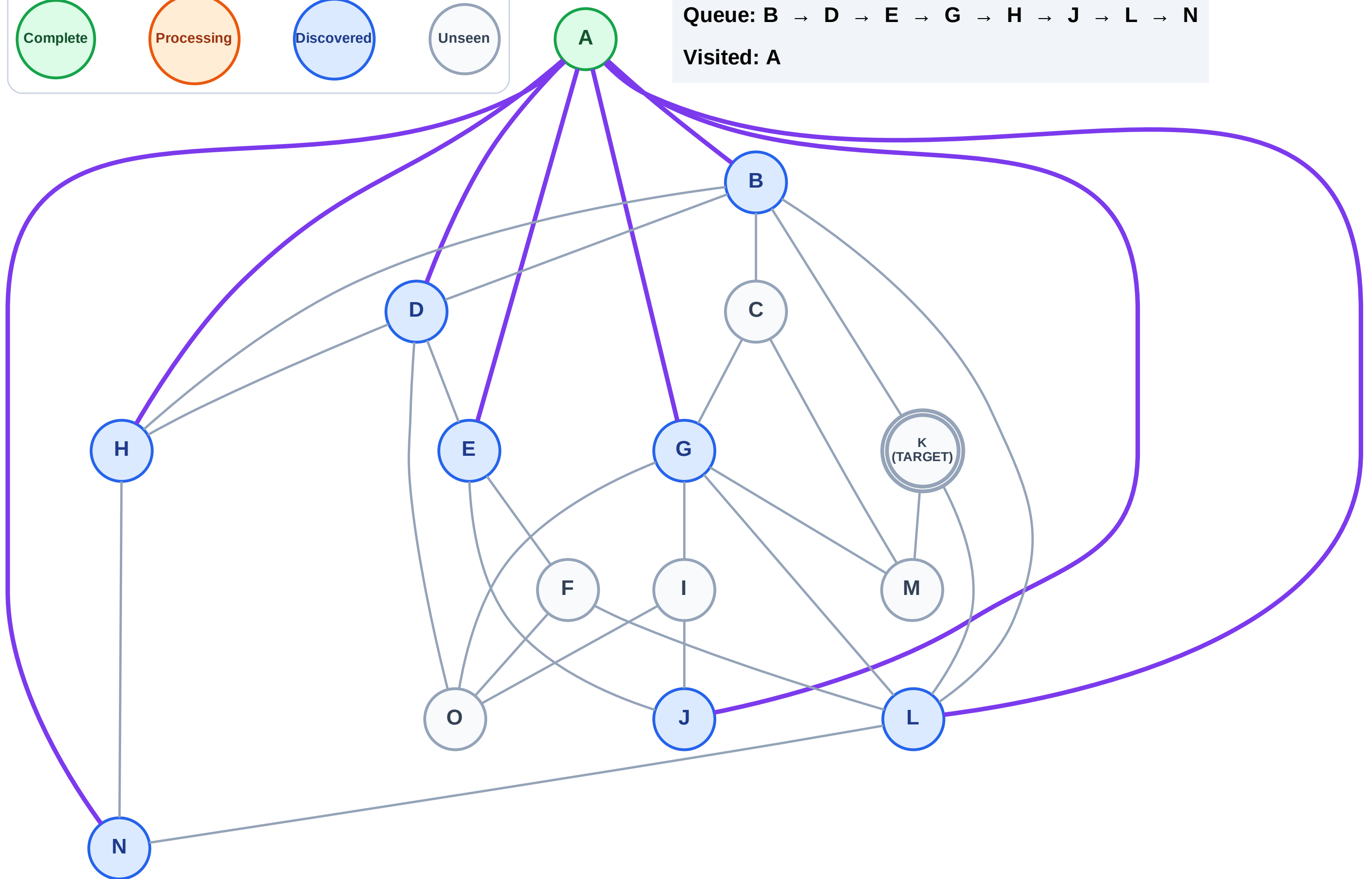
Step 3: Discover B, D, E, G, H, J, L, N from A

Legend



Queue: B → D → E → G → H → J → L → N

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

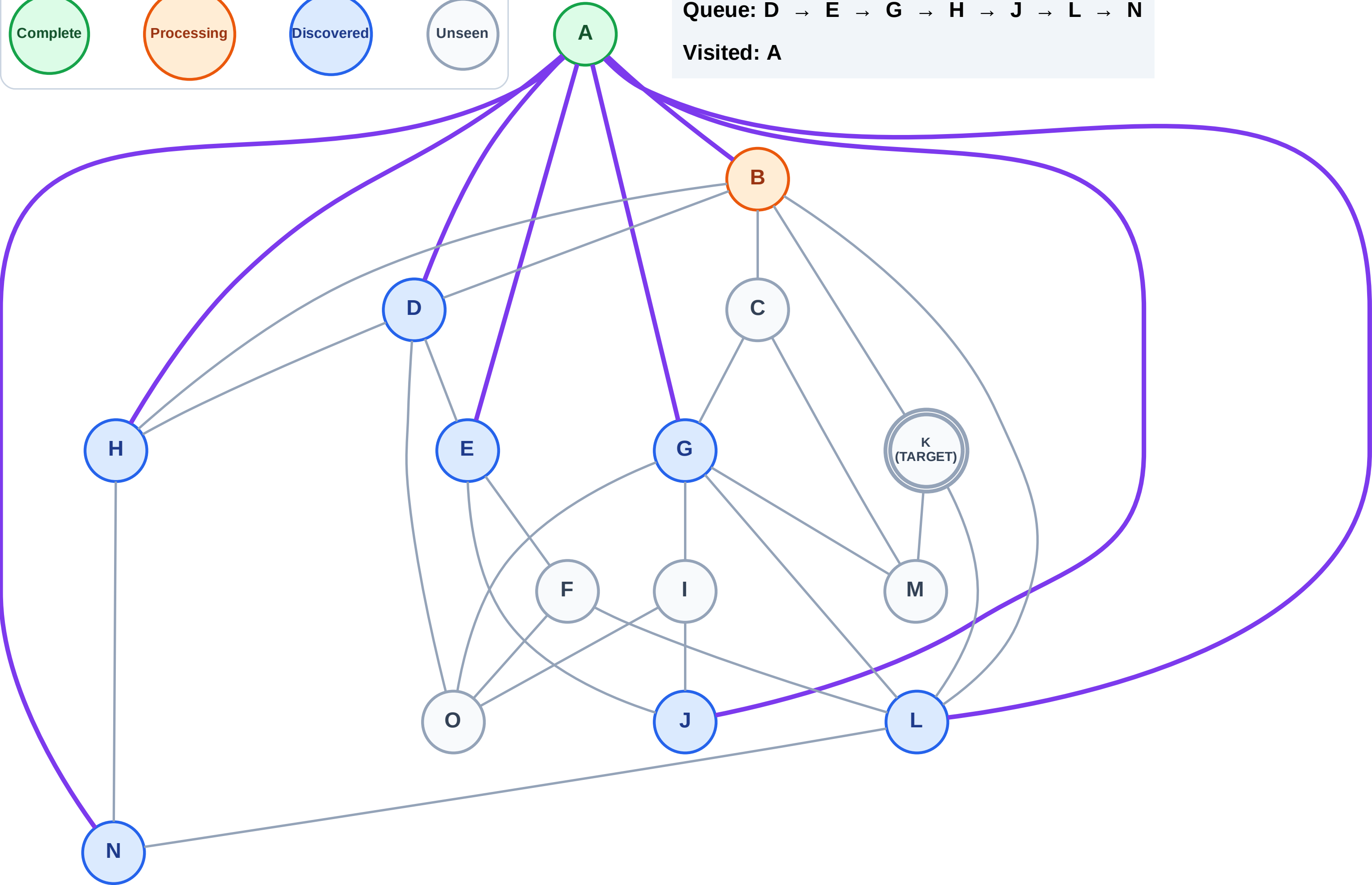
Processing

Discovered

Unseen

Queue: D → E → G → H → J → L → N

Visited: A



BFS Traversal

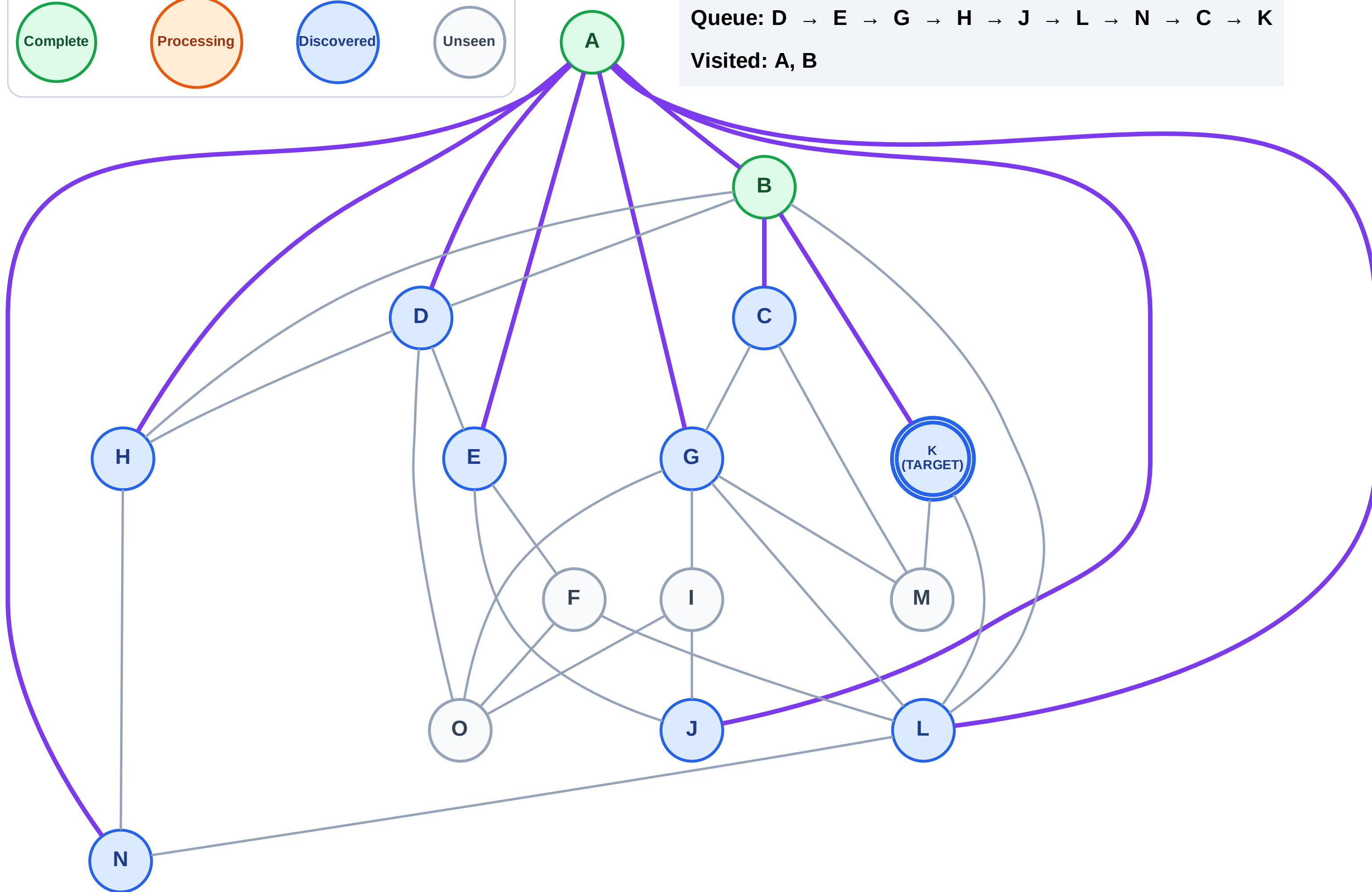
Step 5: Discover C, K from B

Legend



Queue: D → E → G → H → J → L → N → C → K

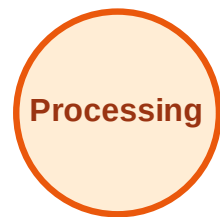
Visited: A, B



BFS Traversal

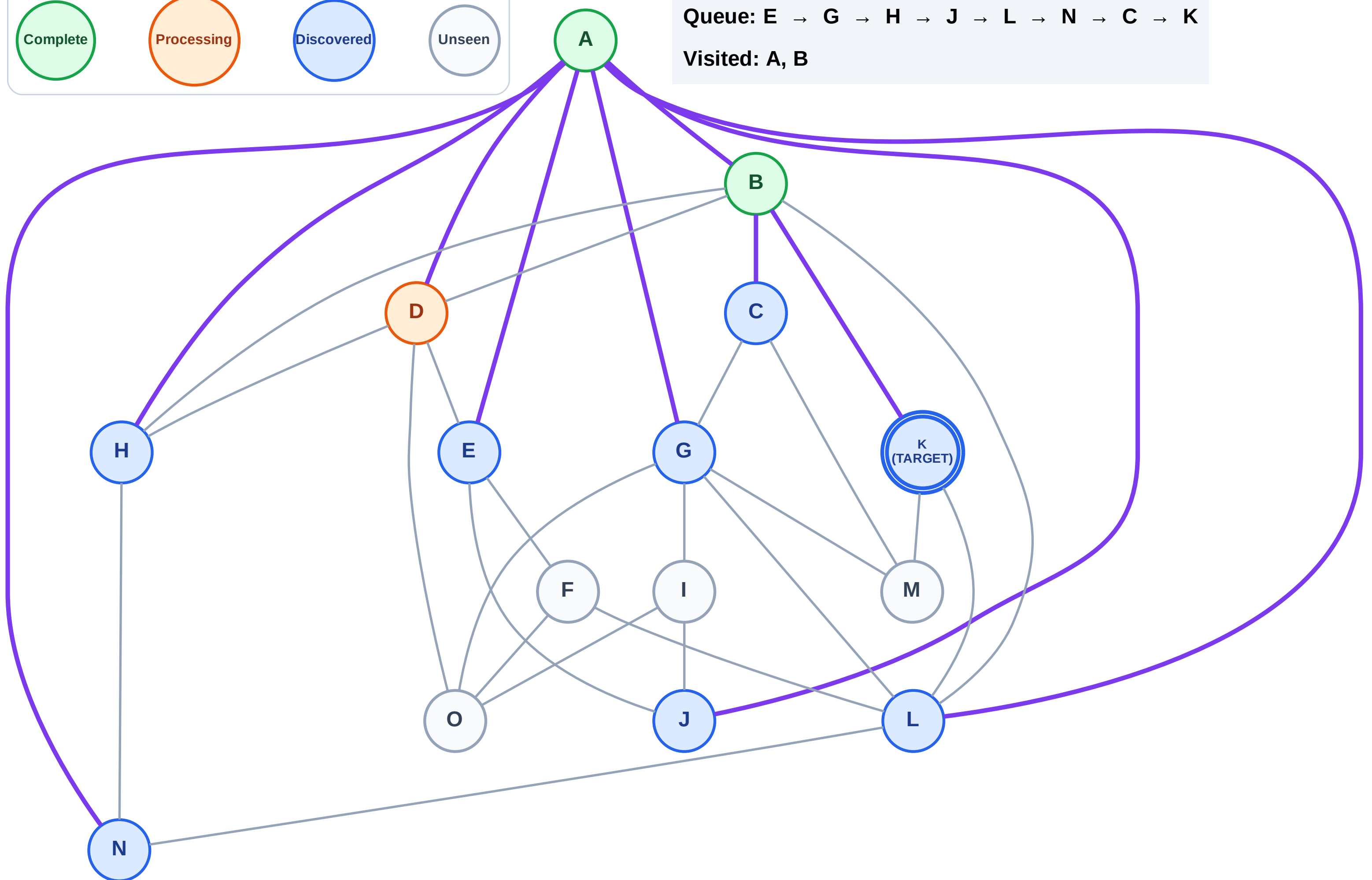
Step 6: Process node D

Legend



Queue: E → G → H → J → L → N → C → K

Visited: A, B



BFS Traversal

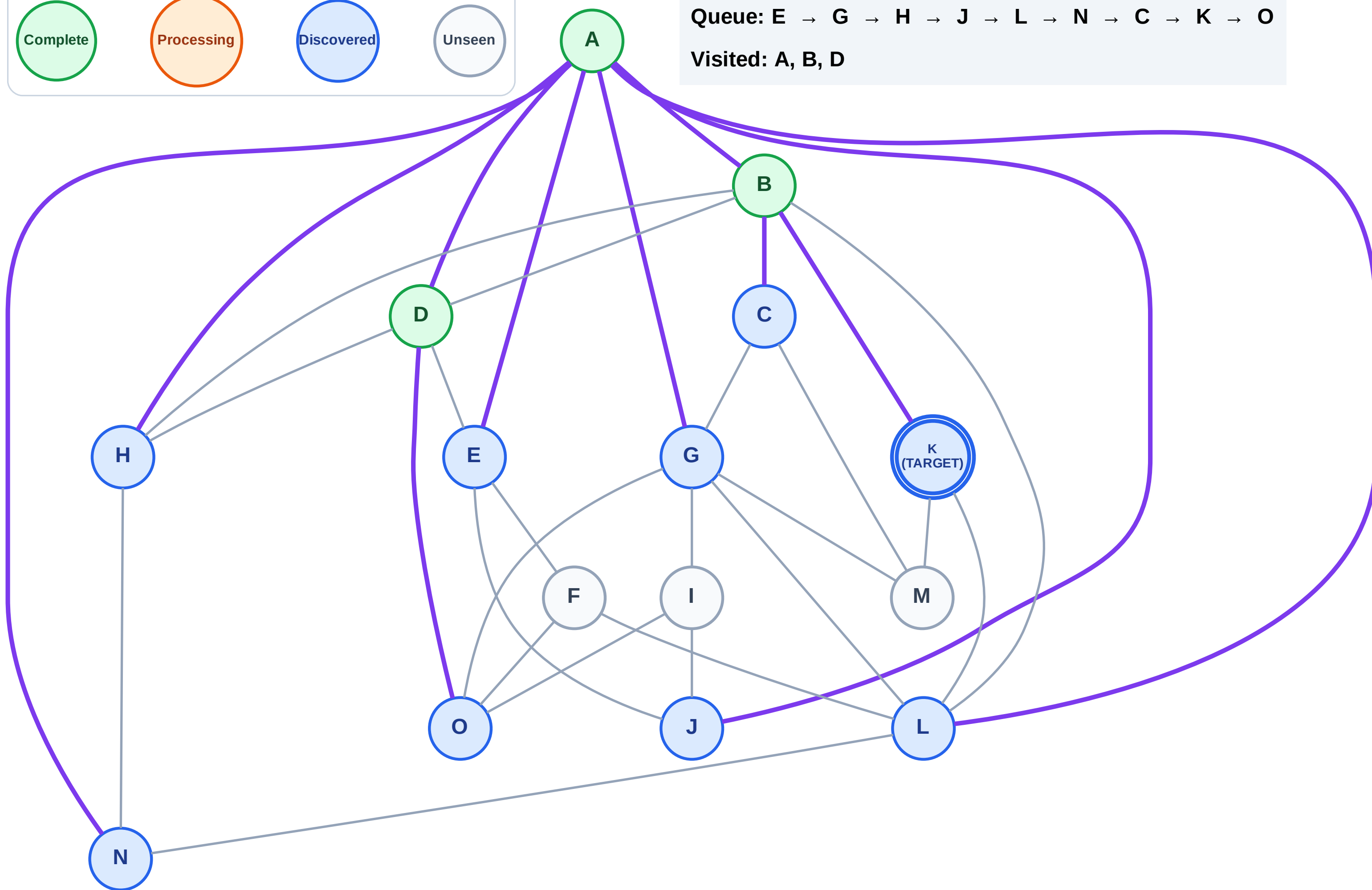
Step 7: Discover O from D

Legend



Queue: E → G → H → J → L → N → C → K → O

Visited: A, B, D



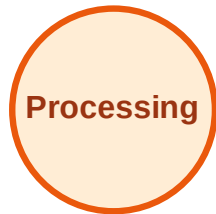
BFS Traversal

Step 8: Process node E

Legend



Complete



Processing



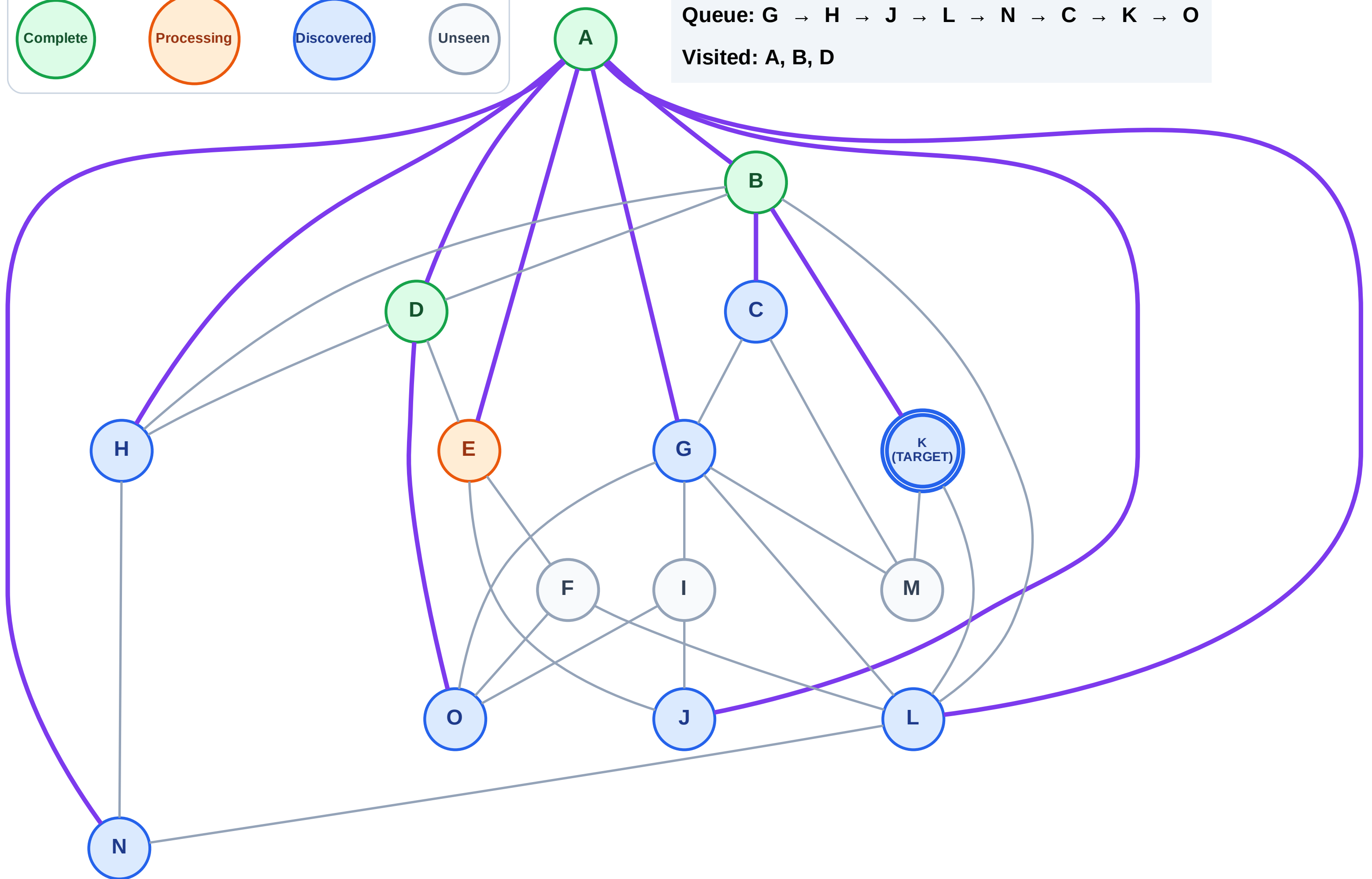
Discovered



Unseen

Queue: G → H → J → L → N → C → K → O

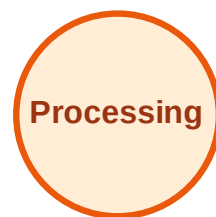
Visited: A, B, D



BFS Traversal

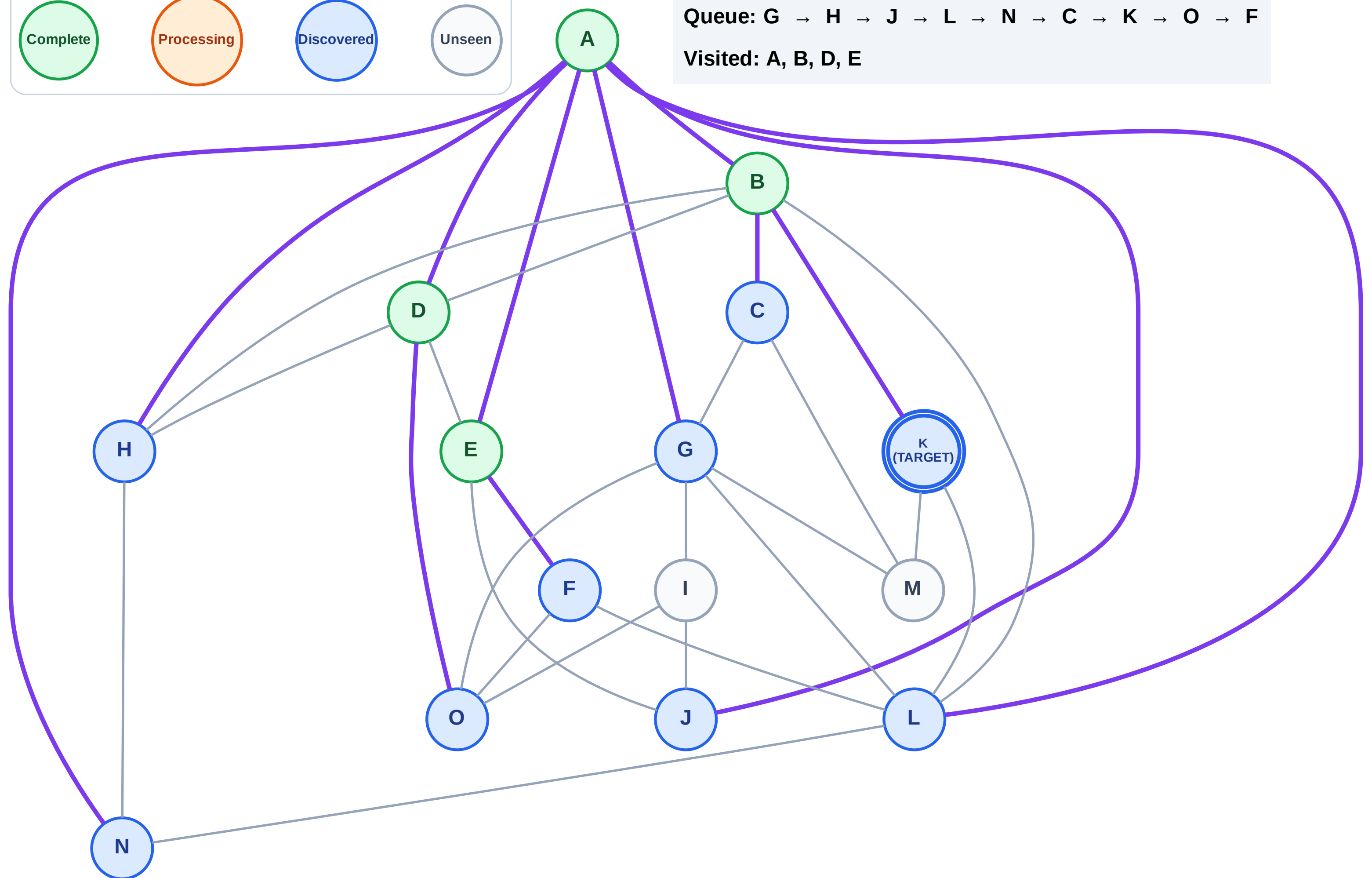
Step 9: Discover F from E

Legend



Queue: G → H → J → L → N → C → K → O → F

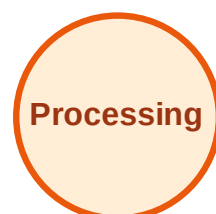
Visited: A, B, D, E



BFS Traversal

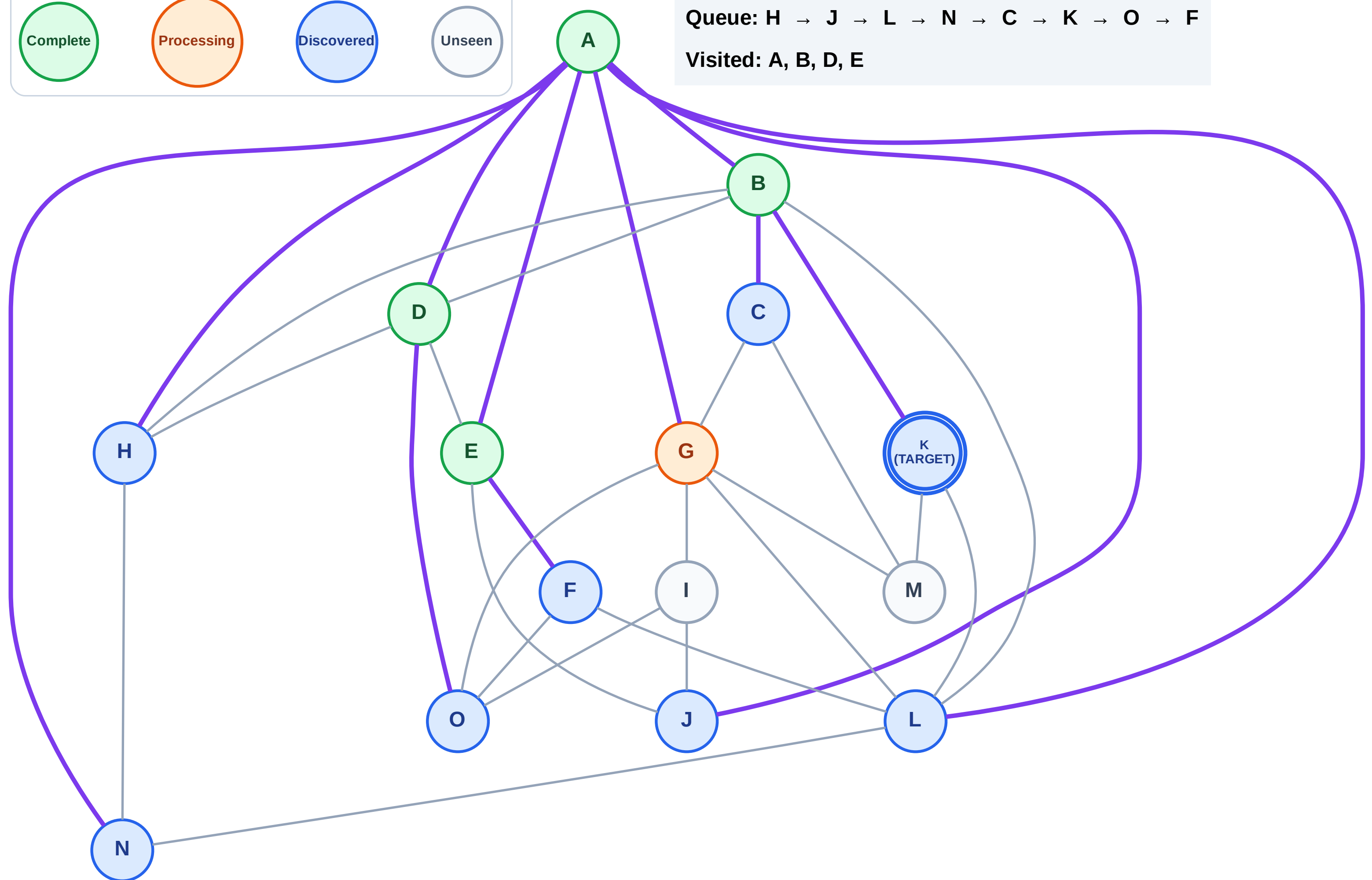
Step 10: Process node G

Legend



Queue: H → J → L → N → C → K → O → F

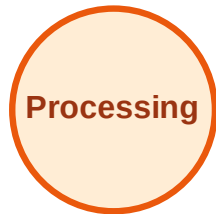
Visited: A, B, D, E



BFS Traversal

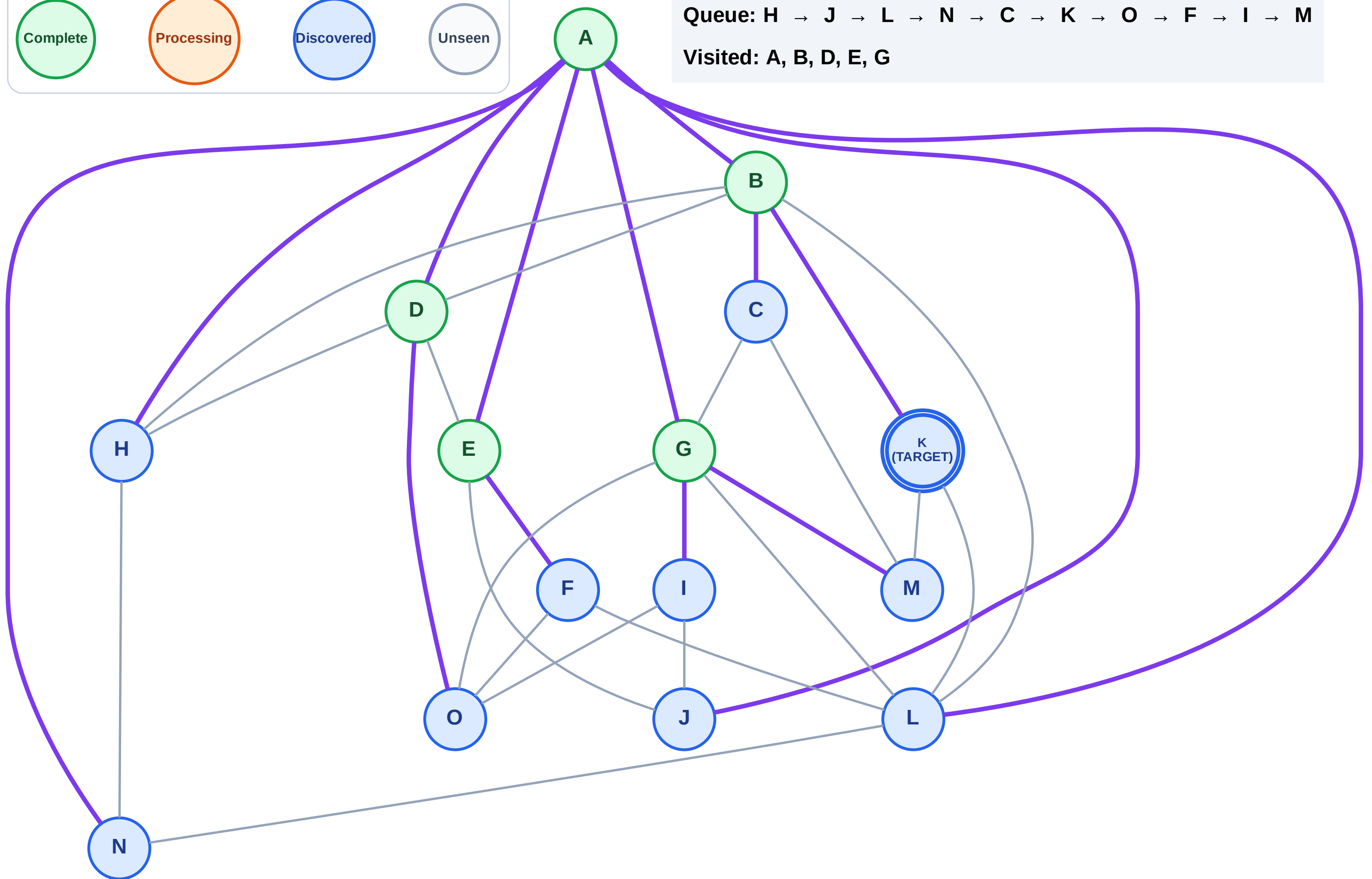
Step 11: Discover I, M from G

Legend



Queue: H → J → L → N → C → K → O → F → I → M

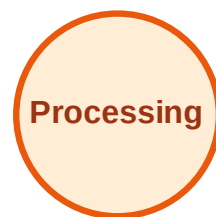
Visited: A, B, D, E, G



BFS Traversal

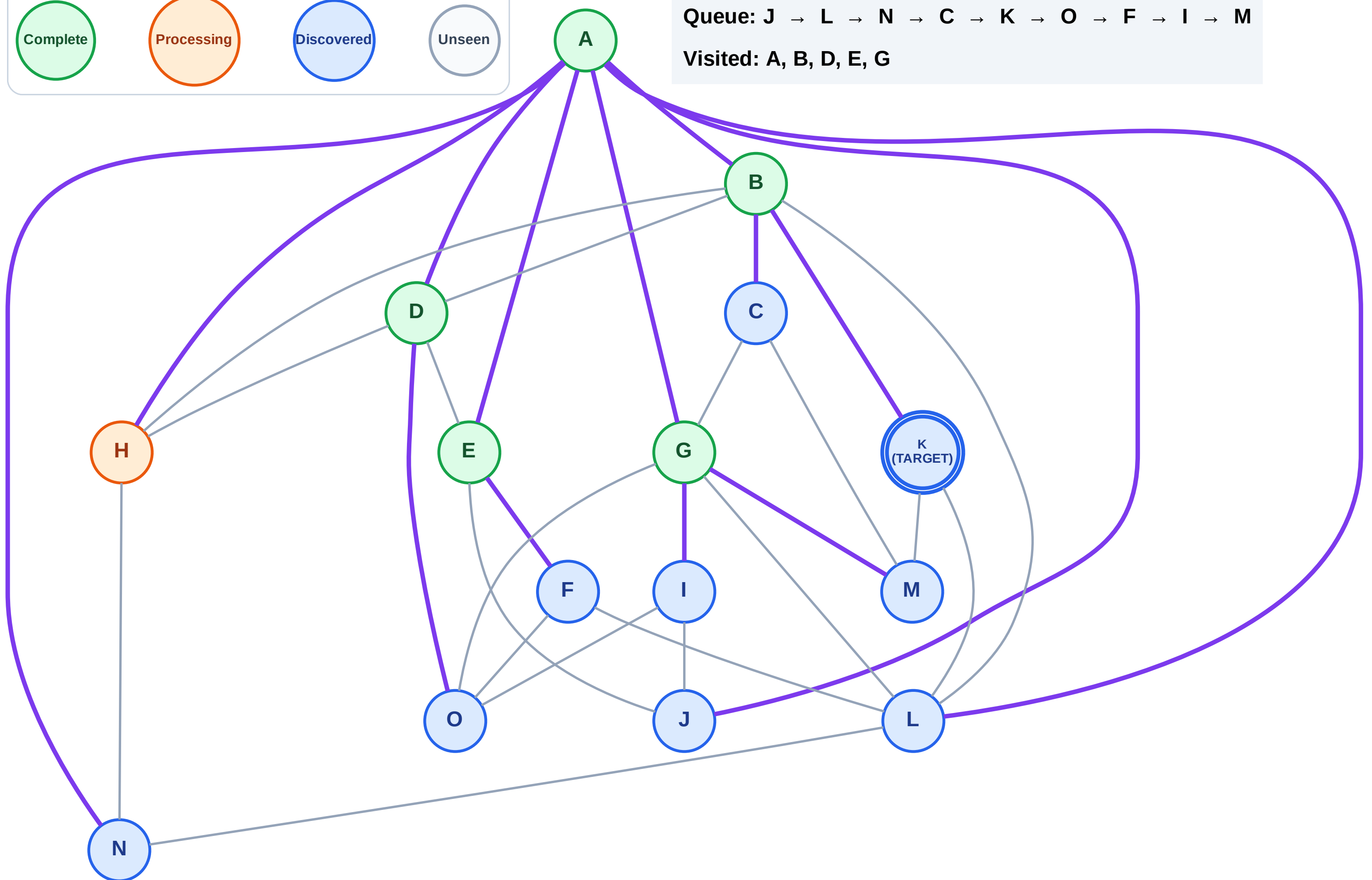
Step 12: Process node H

Legend



Queue: J → L → N → C → K → O → F → I → M

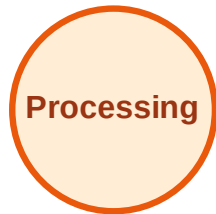
Visited: A, B, D, E, G



BFS Traversal

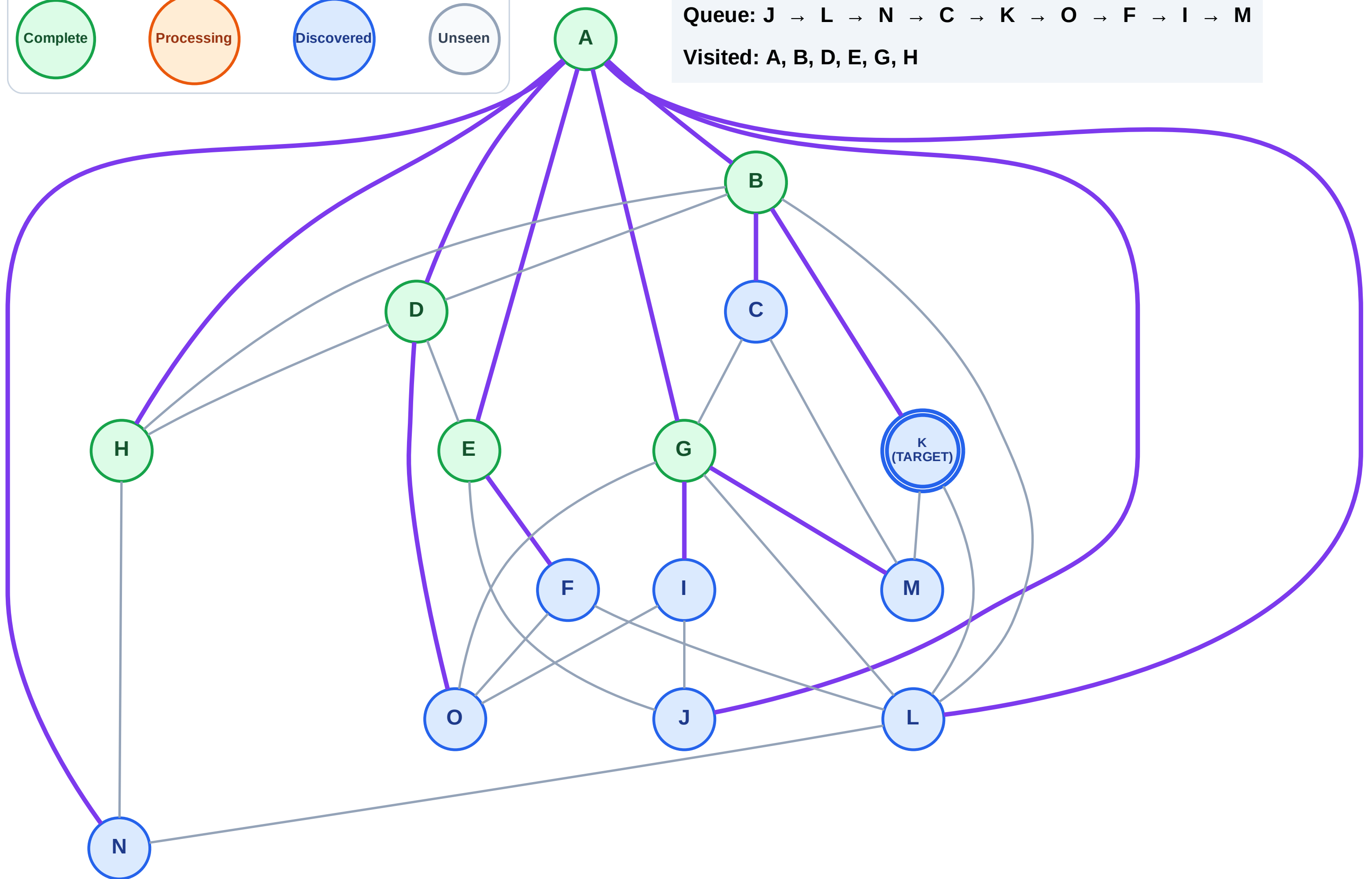
Step 13: No new neighbors from H

Legend



Queue: J → L → N → C → K → O → F → I → M

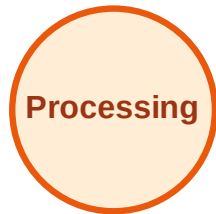
Visited: A, B, D, E, G, H



BFS Traversal

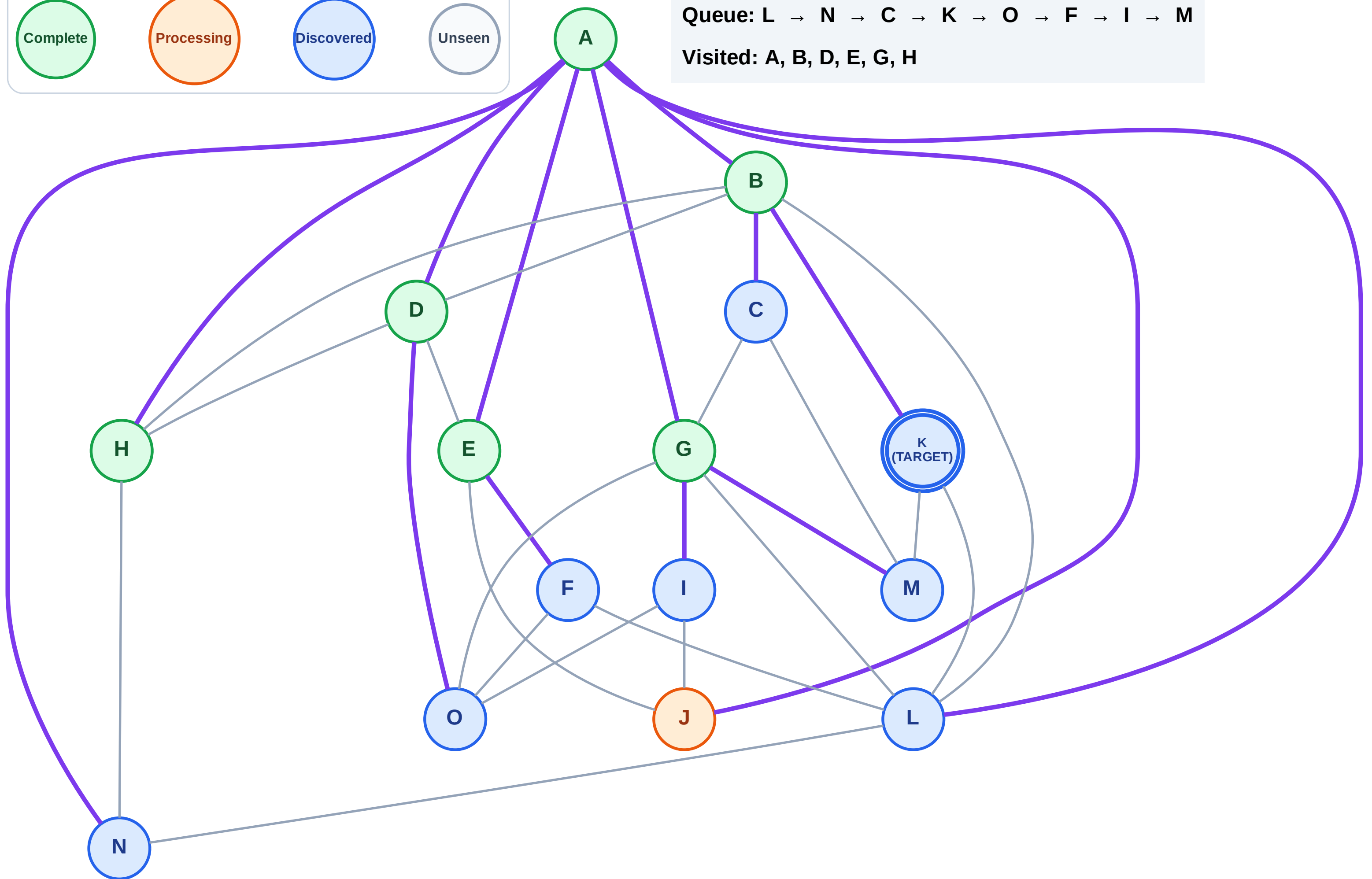
Step 14: Process node J

Legend



Queue: L → N → C → K → O → F → I → M

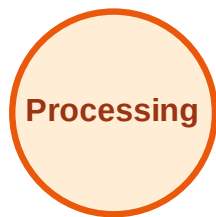
Visited: A, B, D, E, G, H



BFS Traversal

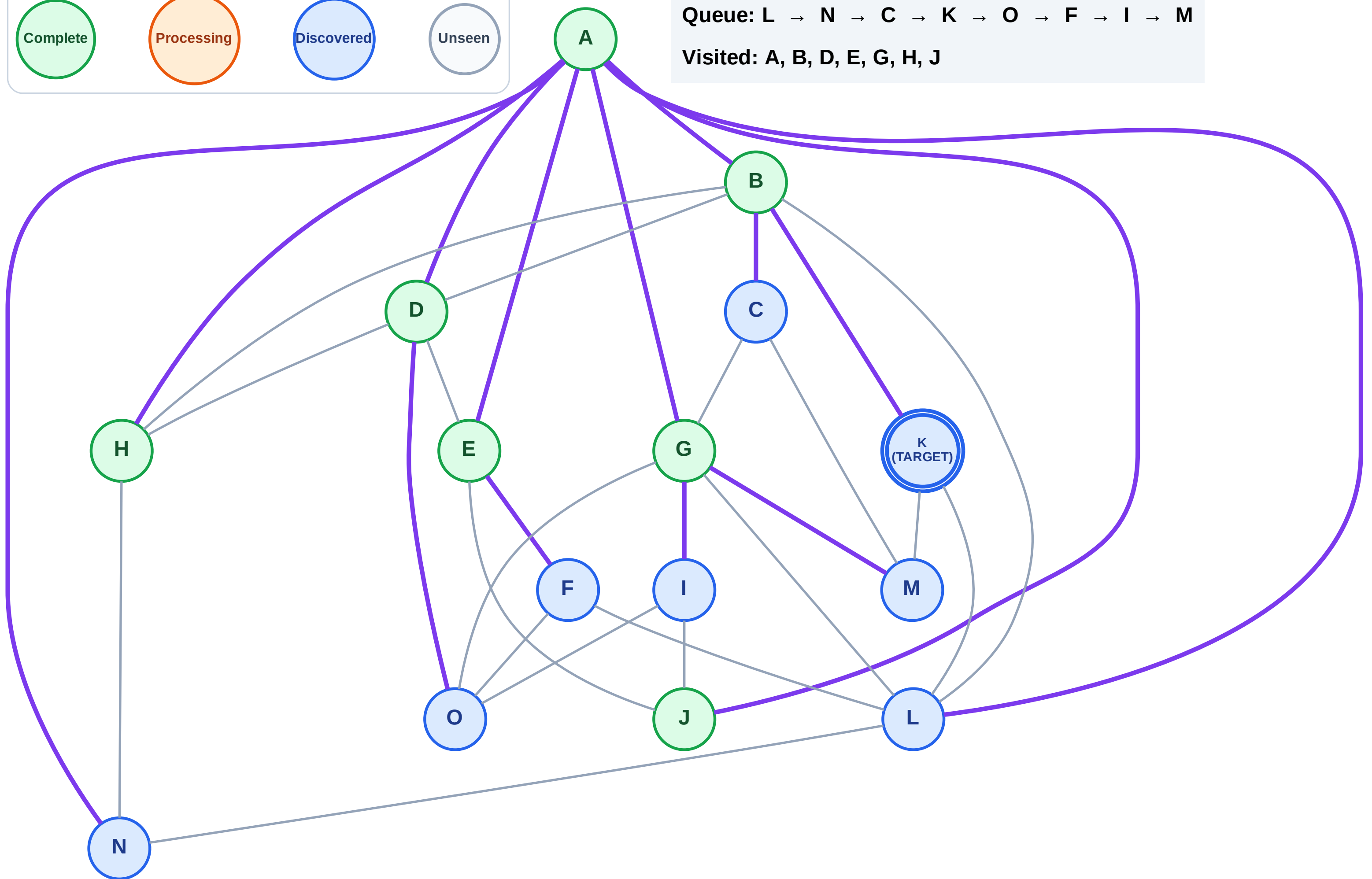
Step 15: No new neighbors from J

Legend



Queue: L → N → C → K → O → F → I → M

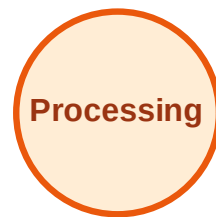
Visited: A, B, D, E, G, H, J



BFS Traversal

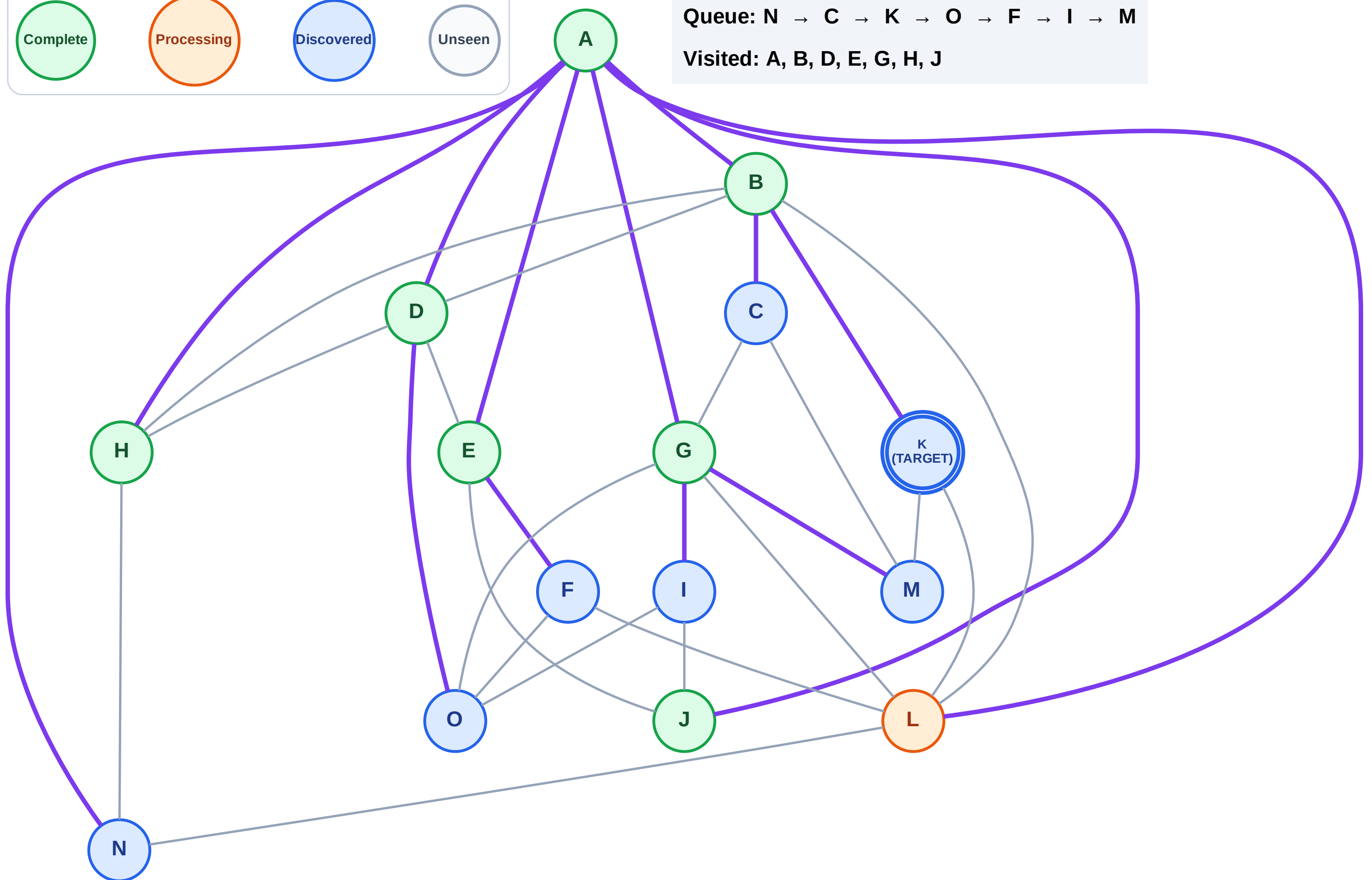
Step 16: Process node L

Legend



Queue: N → C → K → O → F → I → M

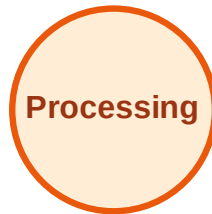
Visited: A, B, D, E, G, H, J



BFS Traversal

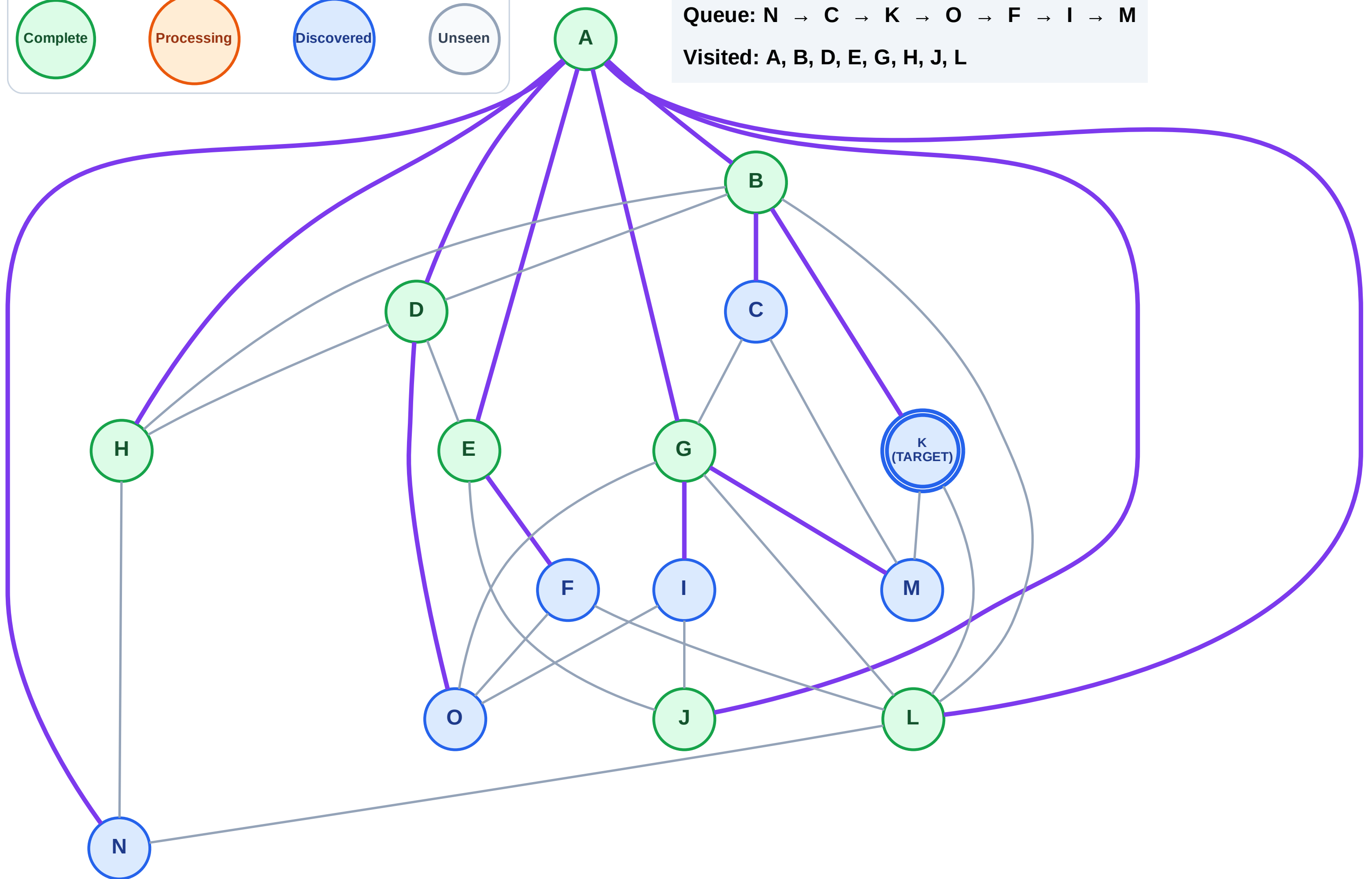
Step 17: No new neighbors from L

Legend



Queue: N → C → K → O → F → I → M

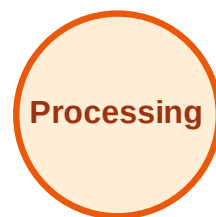
Visited: A, B, D, E, G, H, J, L



BFS Traversal

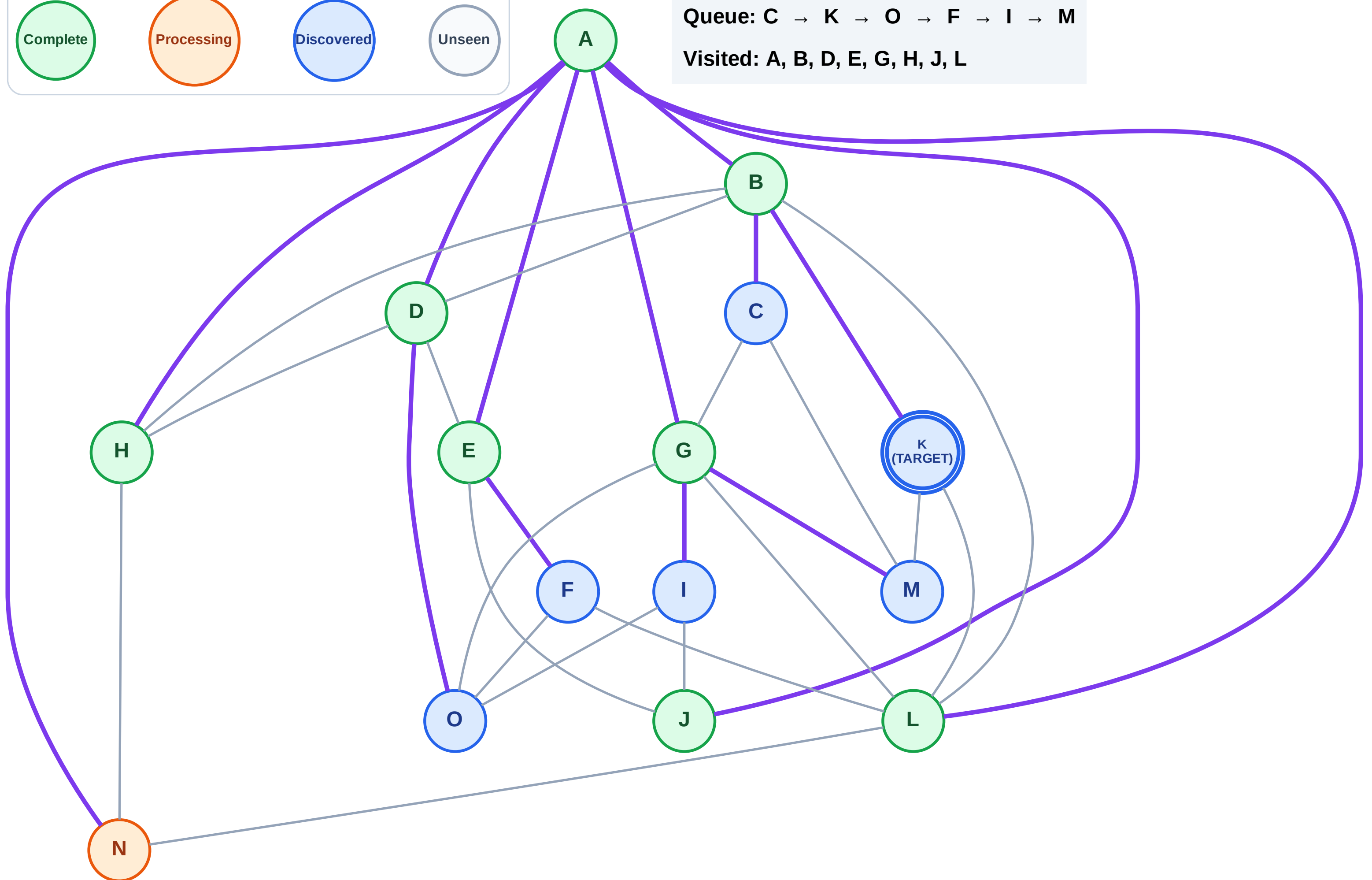
Step 18: Process node N

Legend

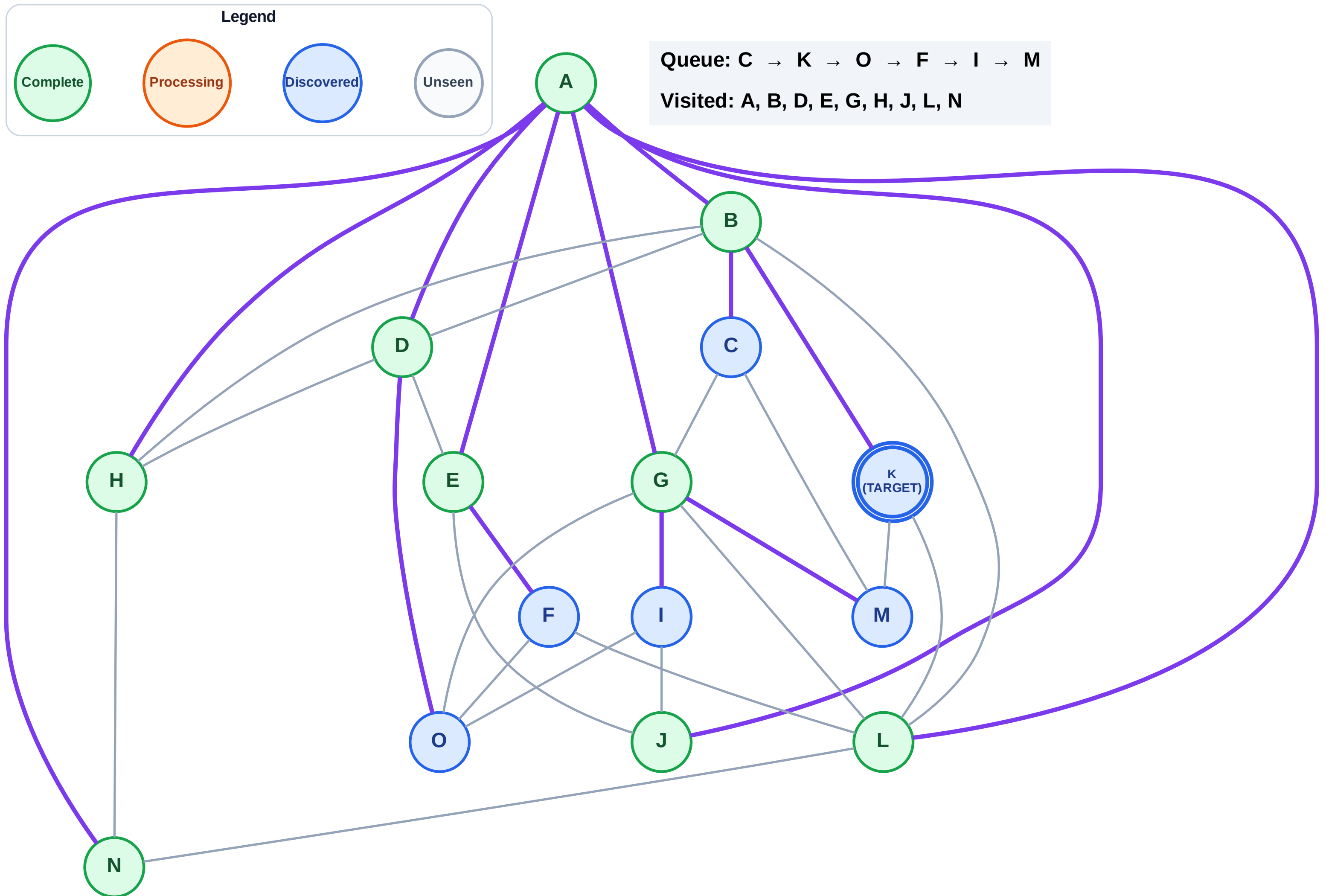


Queue: C → K → O → F → I → M

Visited: A, B, D, E, G, H, J, L



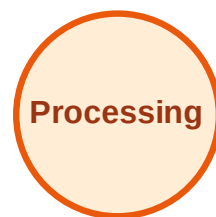
Step 19: No new neighbors from N



BFS Traversal

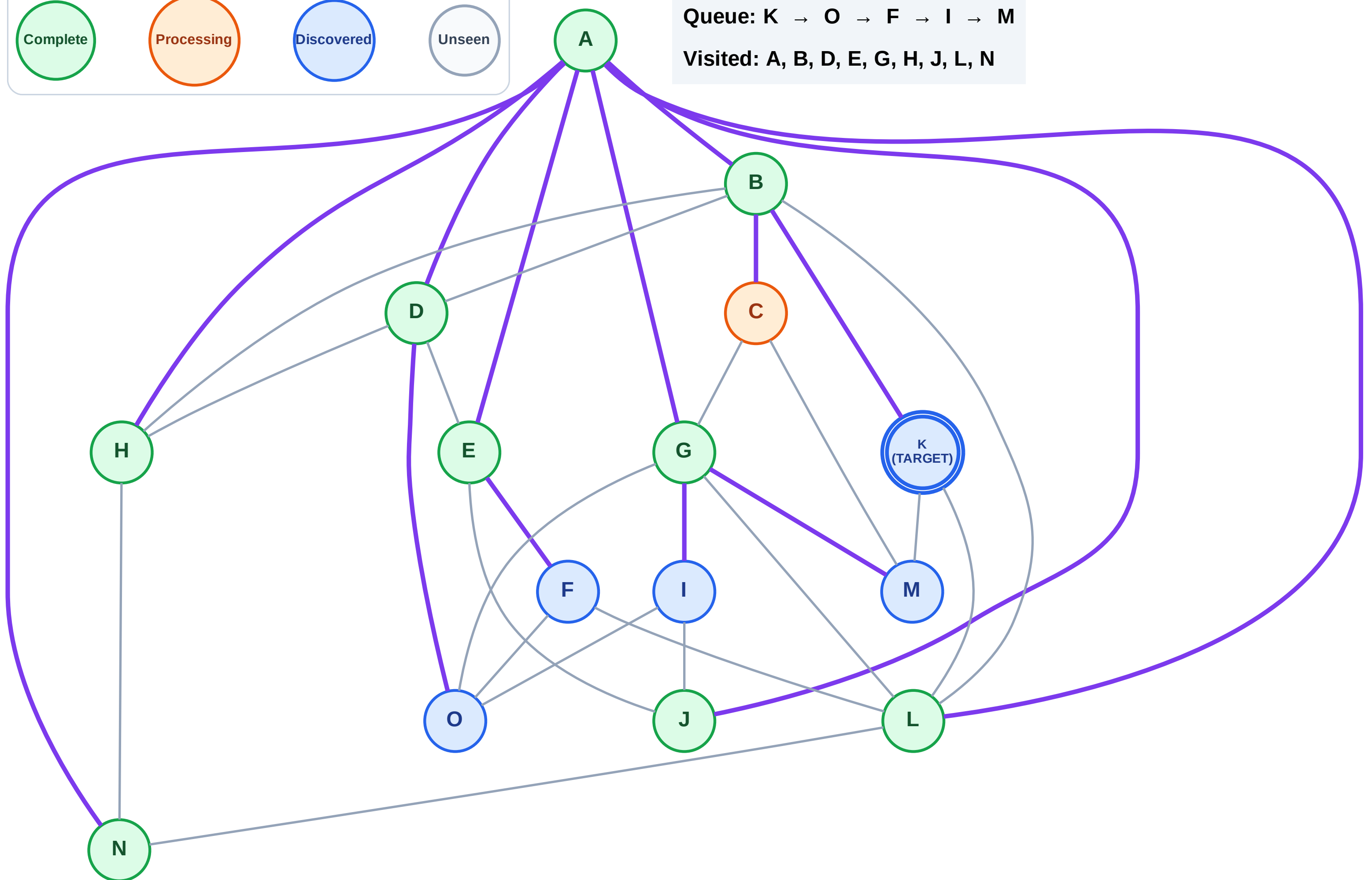
Step 20: Process node C

Legend



Queue: K → O → F → I → M

Visited: A, B, D, E, G, H, J, L, N



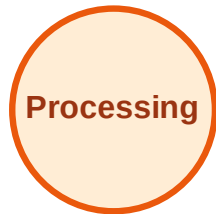
BFS Traversal

Step 21: No new neighbors from C

Legend



Complete



Processing



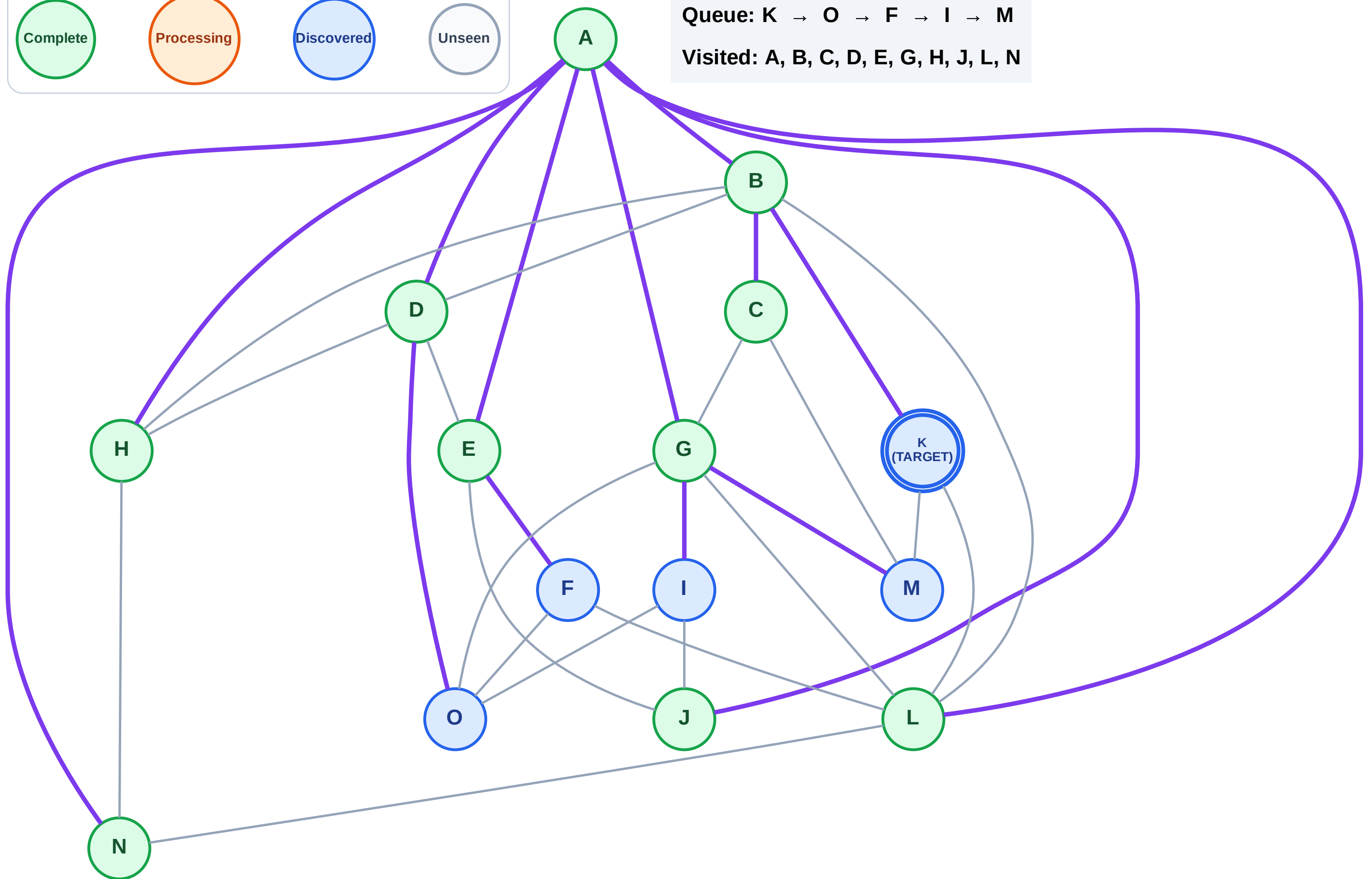
Discovered



Unseen

Queue: K → O → F → I → M

Visited: A, B, C, D, E, G, H, J, L, N



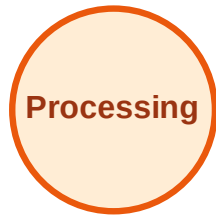
BFS Traversal

Step 22: Process node K

Legend



Complete



Processing



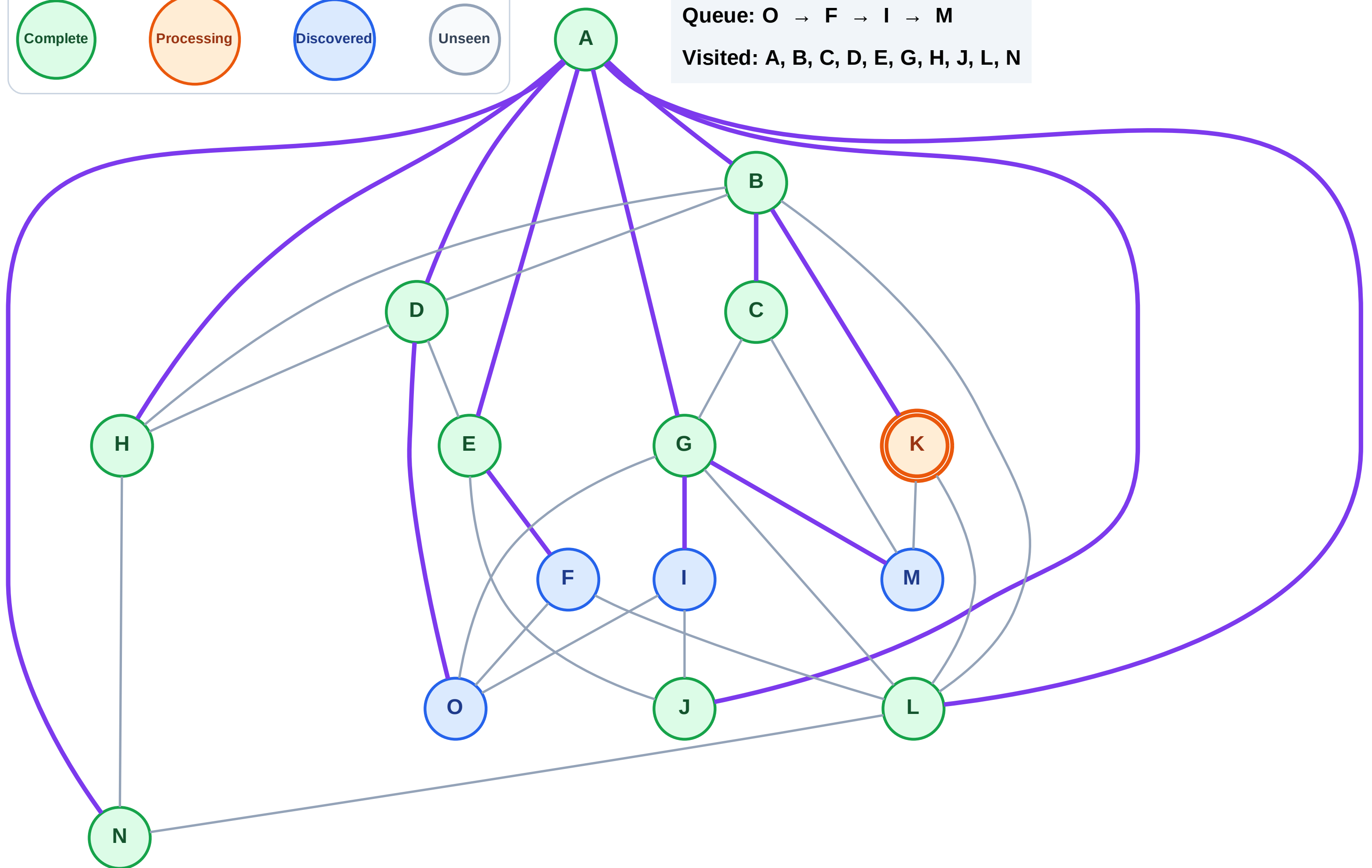
Discovered



Unseen

Queue: O → F → I → M

Visited: A, B, C, D, E, G, H, J, L, N



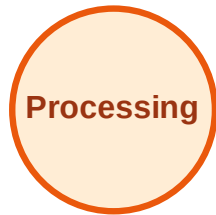
BFS Traversal

Step 23: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Complete



Processing



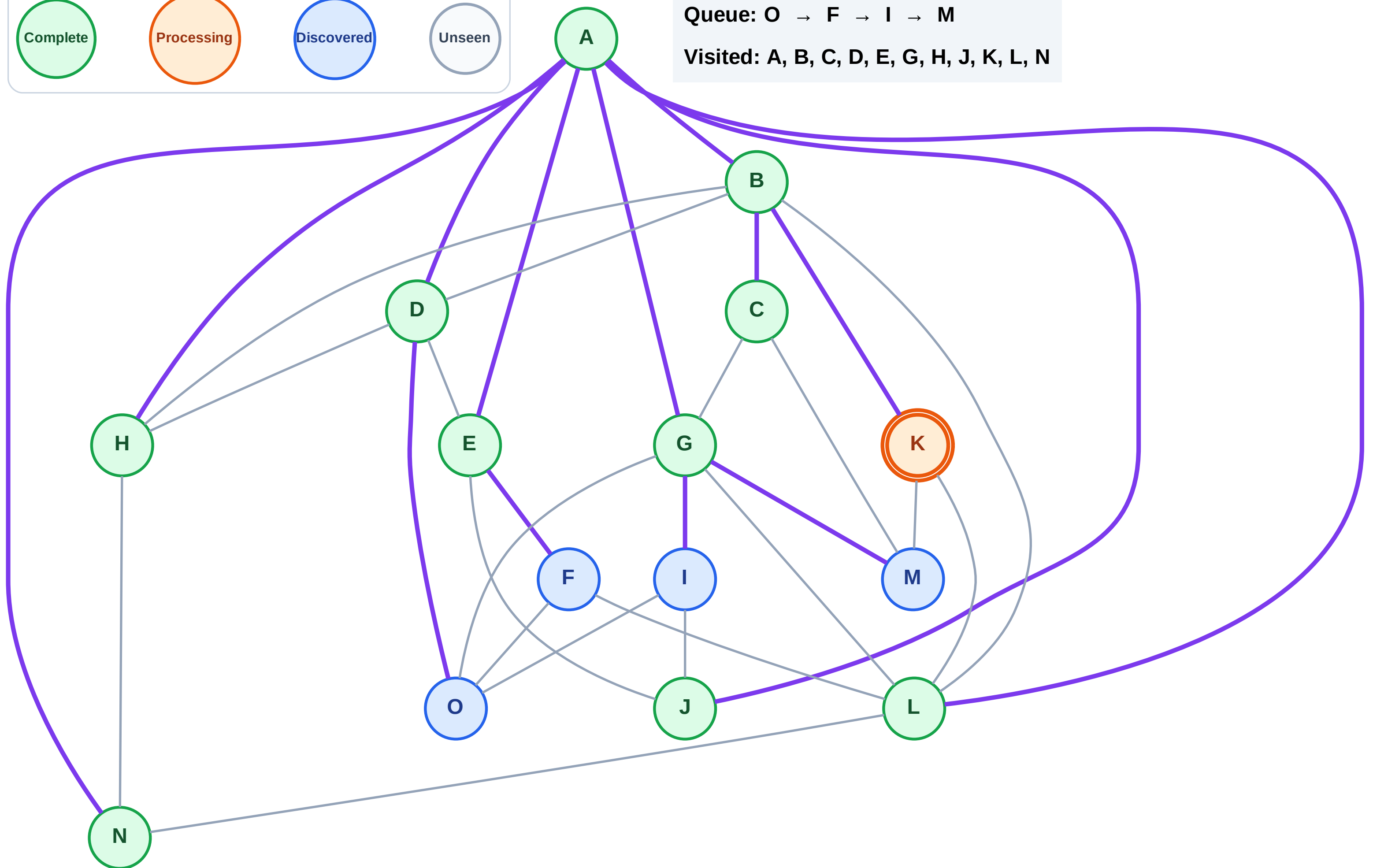
Discovered



Unseen

Queue: O → F → I → M

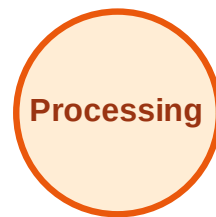
Visited: A, B, C, D, E, G, H, J, K, L, N



DFS Traversal

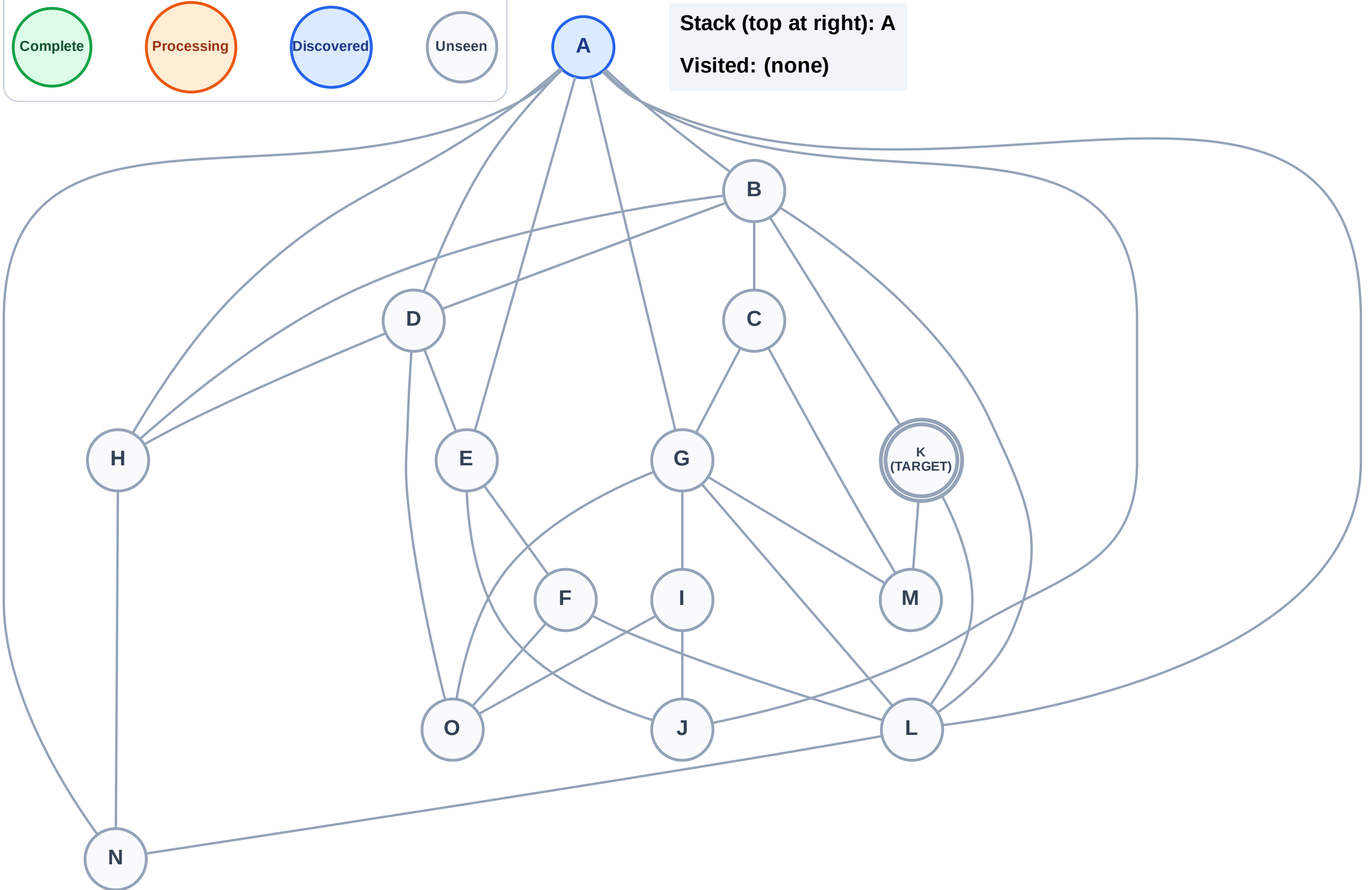
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

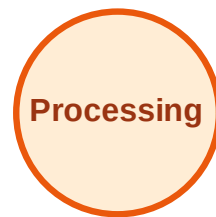
Visited: (none)



DFS Traversal

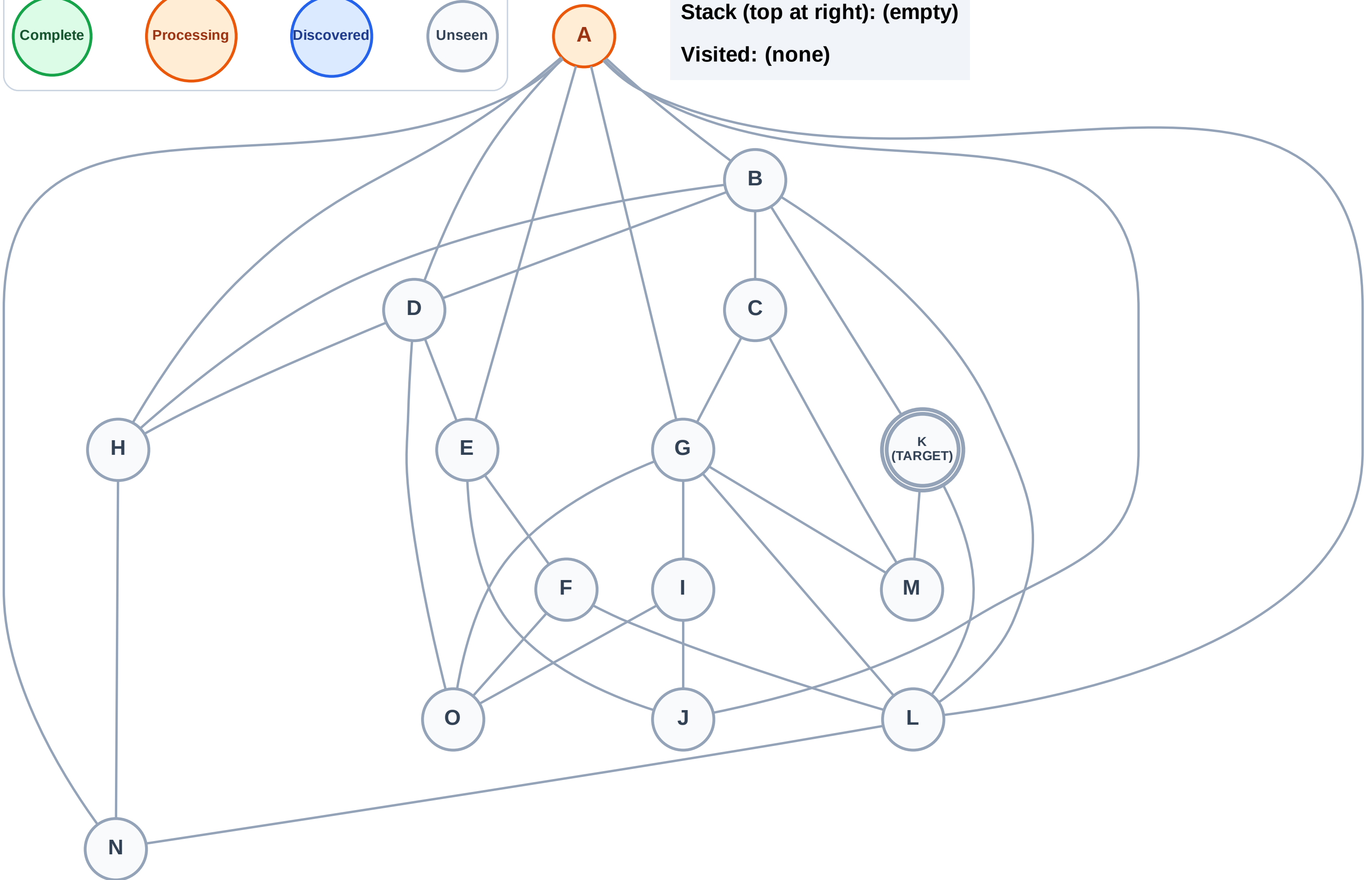
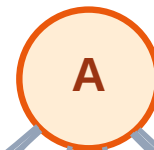
Step 2: Process node A

Legend



Stack (top at right): (empty)

Visited: (none)

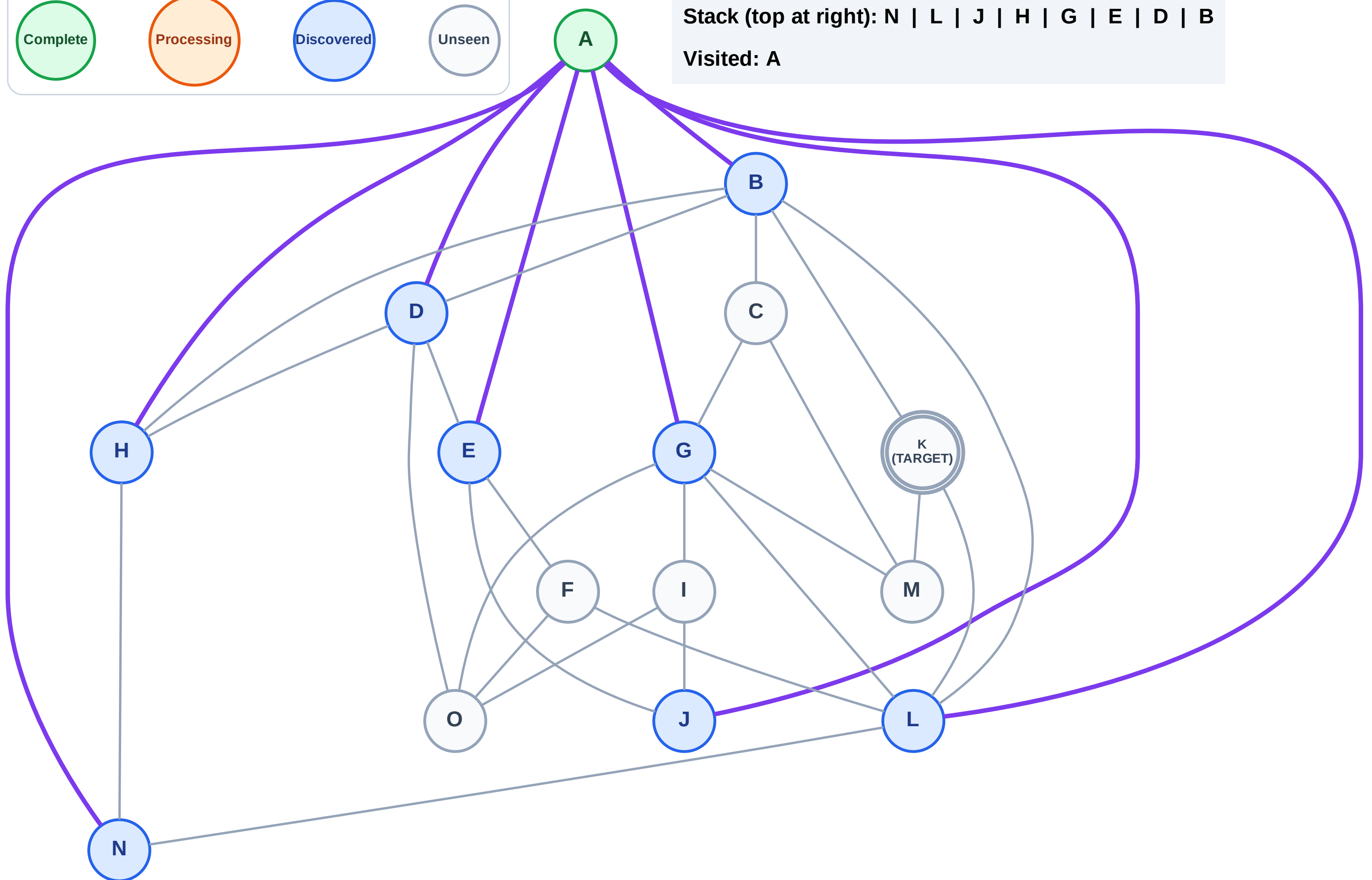


Step 3: Discover N, L, J, H, G, E, D, B from A

Legend

Unseen

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

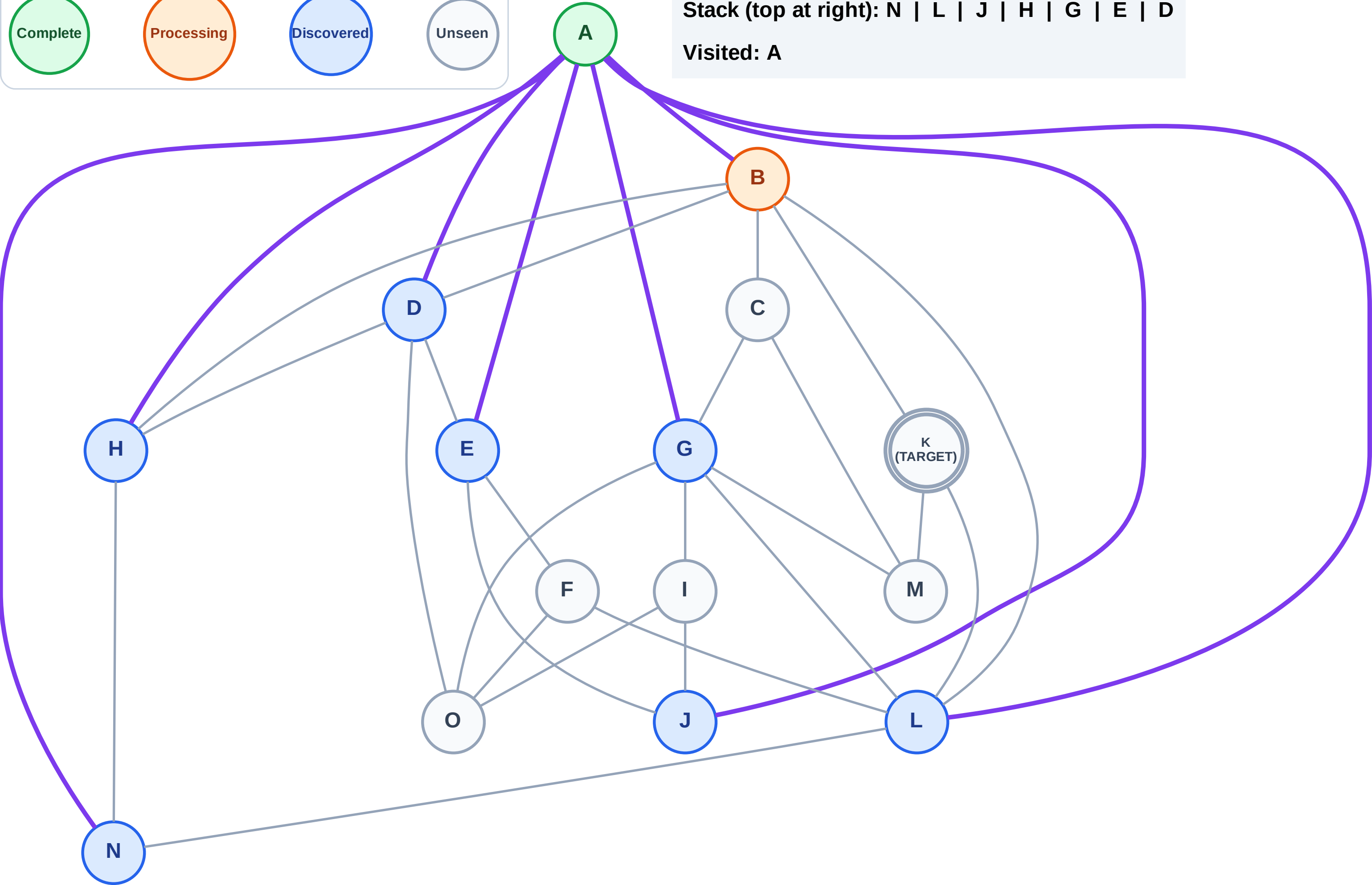
Processing

Discovered

Unseen

Stack (top at right): N | L | J | H | G | E | D

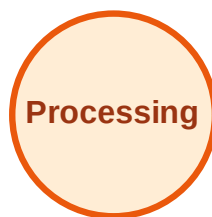
Visited: A



DFS Traversal

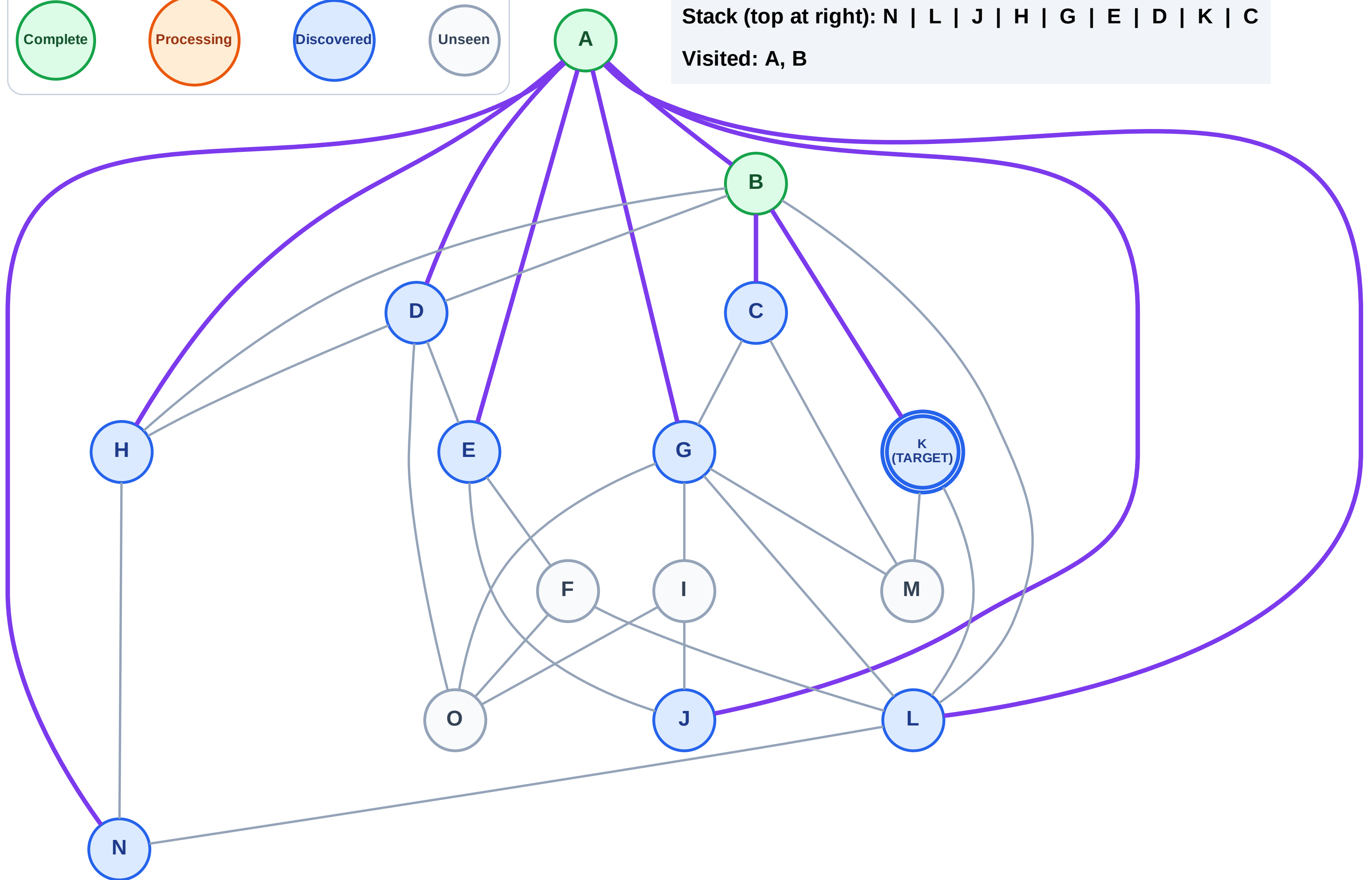
Step 5: Discover K, C from B

Legend



Stack (top at right): N | L | J | H | G | E | D | K | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

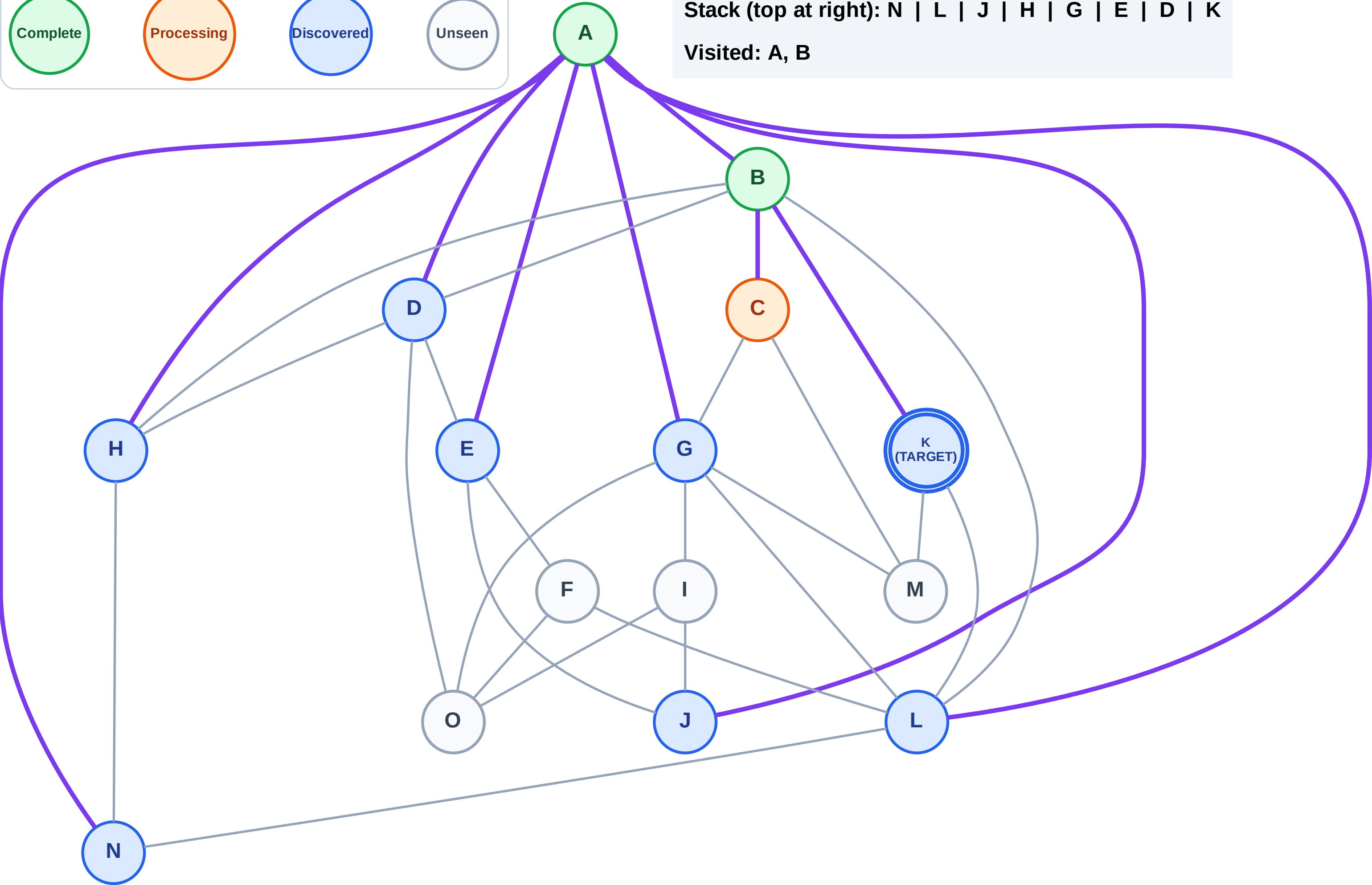
Processing

Discovered

Unseen

Stack (top at right): N | L | J | H | G | E | D | K

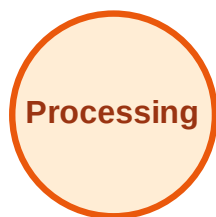
Visited: A, B



DFS Traversal

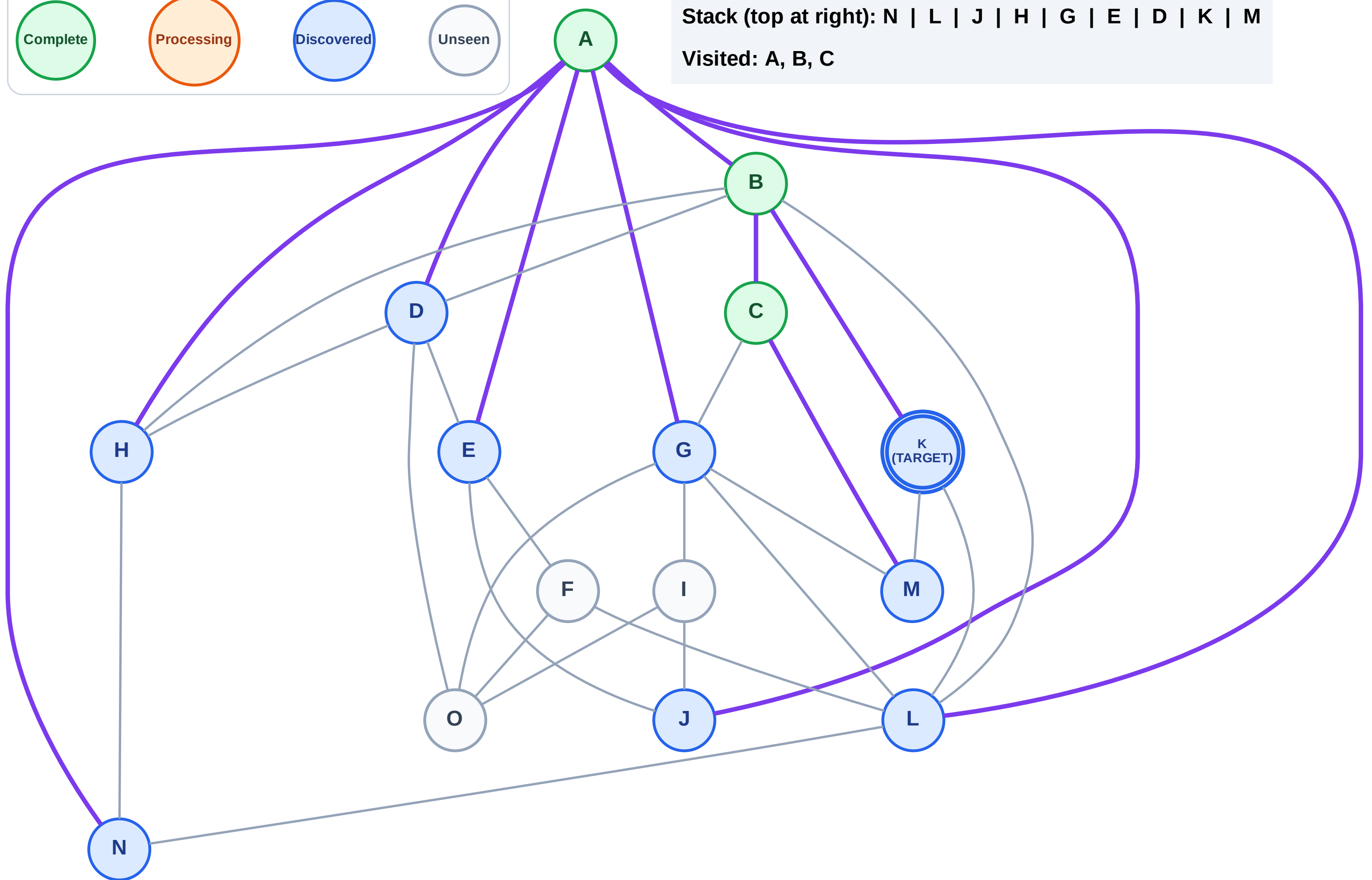
Step 7: Discover M from C

Legend



Stack (top at right): N | L | J | H | G | E | D | K | M

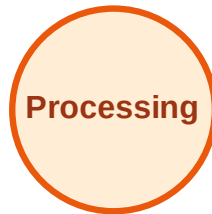
Visited: A, B, C



DFS Traversal

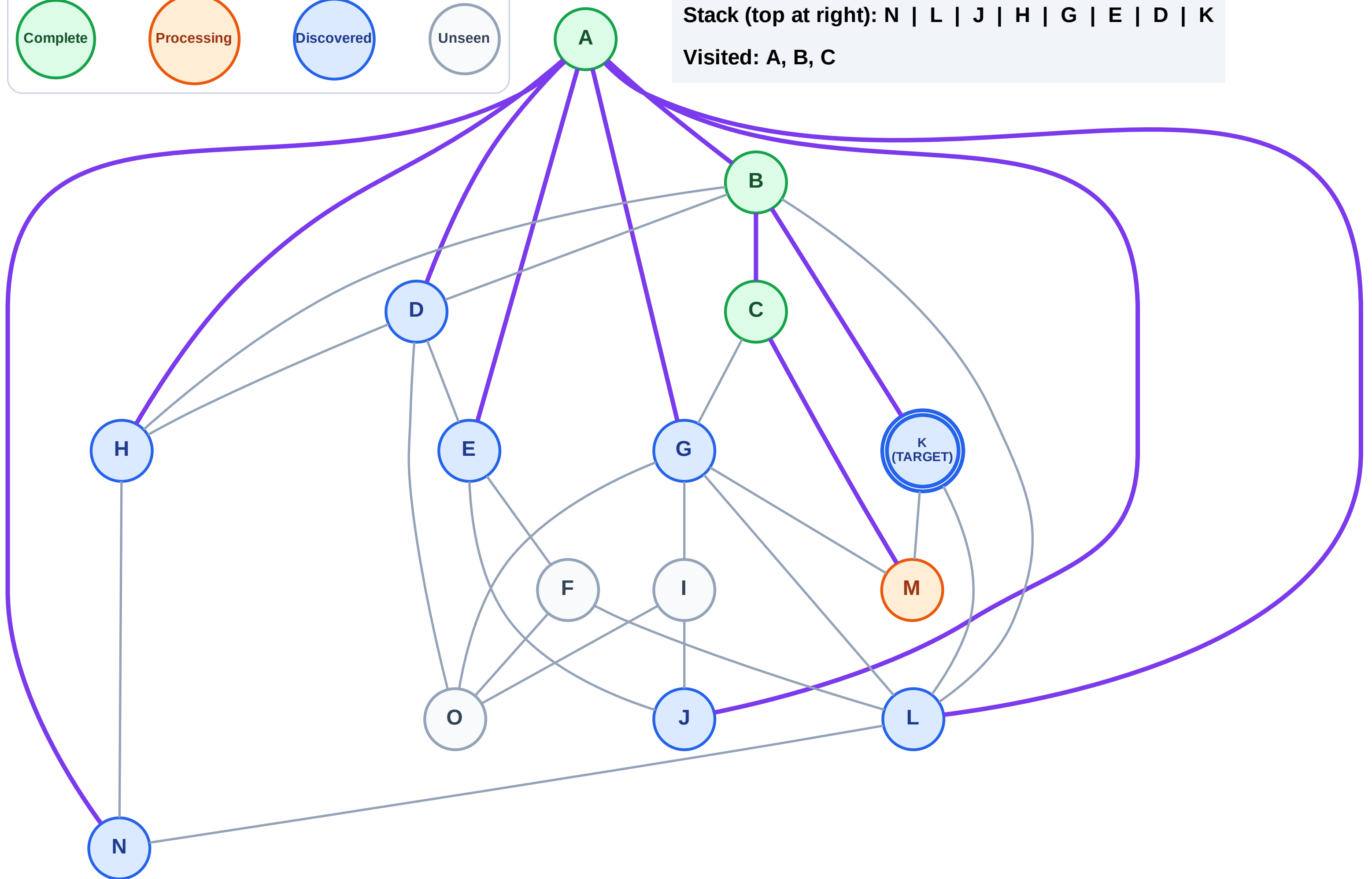
Step 8: Process node M

Legend



Stack (top at right): N | L | J | H | G | E | D | K

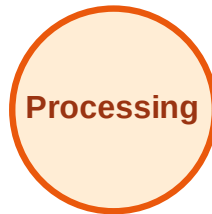
Visited: A, B, C



DFS Traversal

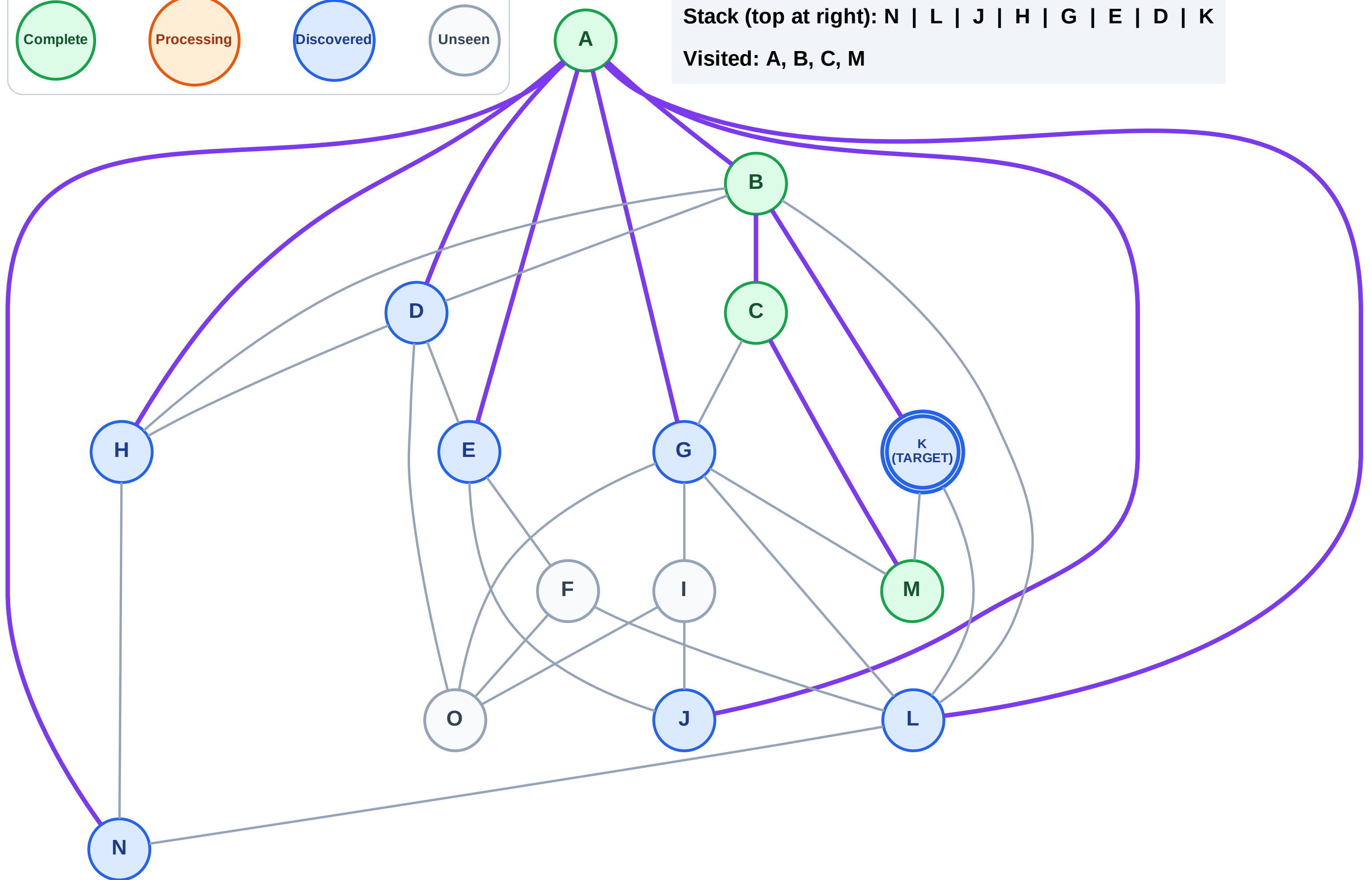
Step 9: No new neighbors from M

Legend



Stack (top at right): N | L | J | H | G | E | D | K

Visited: A, B, C, M



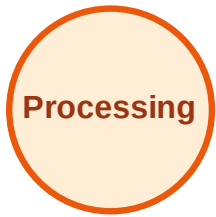
DFS Traversal

Step 10: Process node K

Legend



Complete



Processing



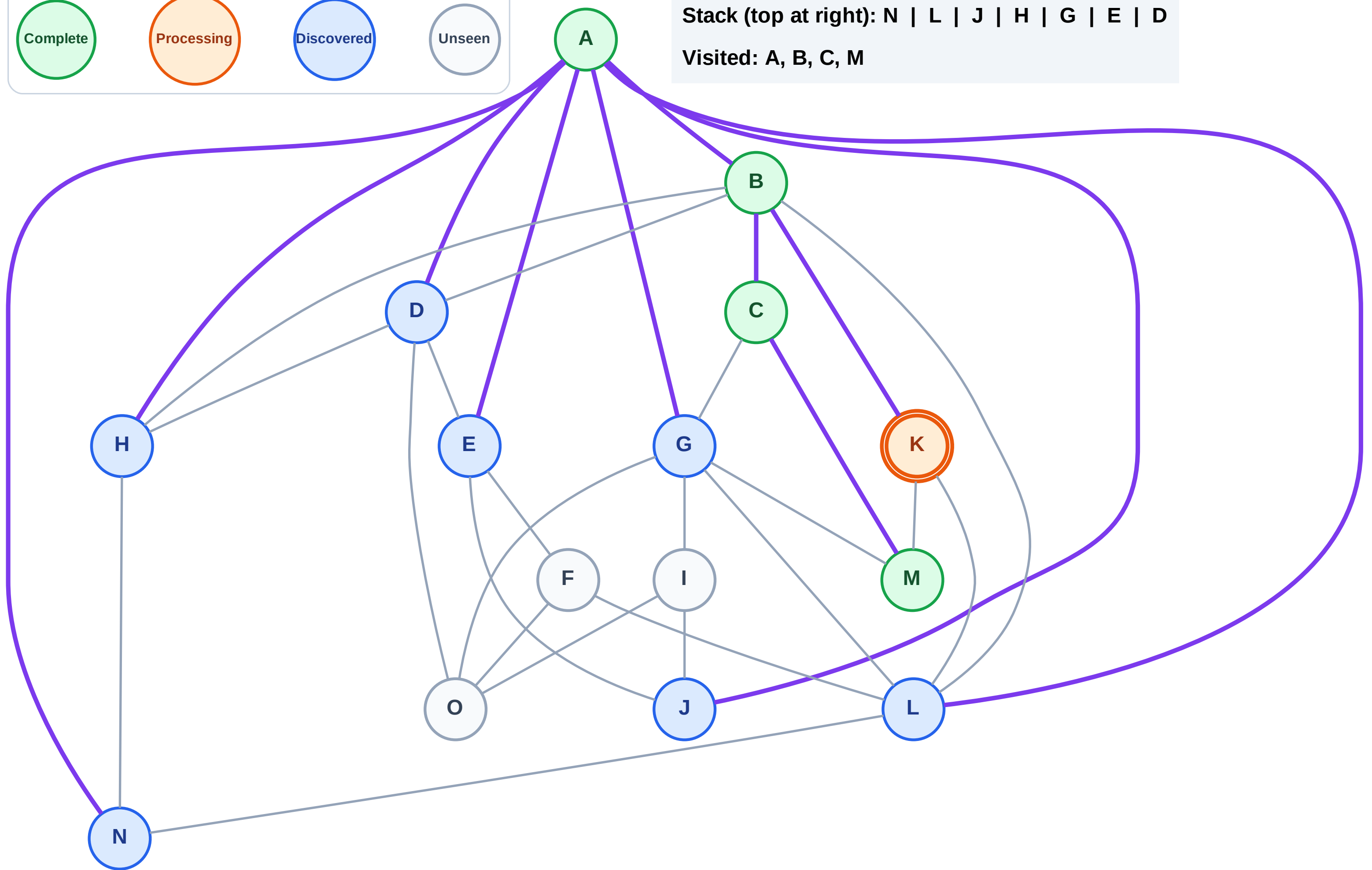
Discovered



Unseen

Stack (top at right): N | L | J | H | G | E | D

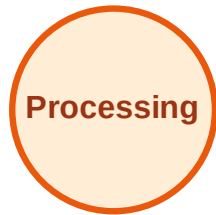
Visited: A, B, C, M



DFS Traversal

Step 11: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Stack (top at right): N | L | J | H | G | E | D

Visited: A, B, C, K, M

